AI-Powered Personalized Learning Platform (Qubo)

By

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A project report submitted to the
Faculty of Computing and Information Technology
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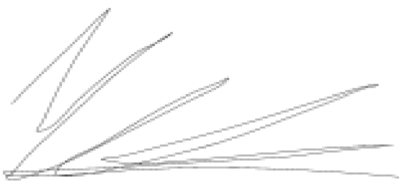
Kuala Lumpur

Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.



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Abstract

Malaysian SPM students often face challenges in identifying learning gaps and optimizing study strategies, leading to inefficient preparation and academic underperformance. This project proposes a Smart Learning Analytics Dashboard that monitors learning patterns and analyzes academic performance using machine learning techniques. The system incorporates spaced repetition algorithms, predictive analytics, and natural language processing to support automated quiz generation from textbook content.

The platform consists of several modules, including performance analytics, personalized study recommendations, content generation, and a lightweight gamification module designed primarily to help students relax and reduce study stress rather than serve as a core learning or practice mechanism. This relaxation-focused mini-game aims to maintain motivation and mental well-being while studying core SPM subjects, particularly those in the science stream.

The system is developed using an Agile software development methodology with iterative testing to ensure reliability, usability, and prediction accuracy. The expected outcome is an intelligent learning companion that assists students in improving engagement, identifying at-risk learners early, and providing data-driven insights to enhance study efficiency. Ultimately, the system aims to improve SPM examination performance and promote self-directed learning among Malaysian secondary school students.

Acknowledgement

First and foremost, we would like to express our deepest gratitude to our project supervisor, Mr. Ten Shai Cheong, for his invaluable guidance, unwavering support, and expert mentorship throughout this journey. His insightful feedback and constant encouragement enabled us to navigate challenges and deepen our understanding of the project's complexities. We are truly thankful for the time and dedication he invested in steering our work to completion.

We are also profoundly grateful to our families for their unconditional support and patience during this intensive period. Their belief in our abilities and their emotional encouragement provided us with the resilience and motivation needed to persevere.

Furthermore, we extend our sincere thanks to our classmates and friends who offered valuable perspectives, engaged in thoughtful discussions, and provided assistance whenever needed. Their camaraderie and collaborative spirit enriched our project experience.

Lastly, I wish to acknowledge my partner—Yong Chao Juin. This project was a true collaborative effort, and our mutual dedication, shared responsibility, and constructive teamwork were fundamental to its successful outcome.

Without the generous contributions and support from all these individuals, our final year project would not have been realized. To each of you, we offer our heartfelt appreciation.

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Chapter 1

Introduction

1 Introduction

The Sijil Pelajaran Malaysia (SPM) examination represents a critical milestone in the Malaysian education system, serving as a gateway to tertiary education and career opportunities. However, students preparing for this high-stakes examination face numerous challenges that impede their learning effectiveness and academic success. Traditional learning approaches often fail to provide the personalized support and real-time feedback necessary for students to identify and address their knowledge gaps effectively.

In today's digital age, artificial intelligence and machine learning technologies offer unprecedented opportunities to transform education through personalization and data-driven insights. Qubo is designed to leverage these technologies to create an AI-powered personalized learning platform specifically tailored for SPM students. The platform aims to address the fundamental challenge of inefficient learning by providing intelligent analytics, personalized study recommendations, and automated content generation tools.

This project introduces a comprehensive learning analytics dashboard that employs spaced repetition algorithms, predictive analytics, and natural language processing to help students maximize their learning efficiency. By tracking performance patterns and analyzing learning behaviors, Qubo empowers students to take control of their academic journey and achieve better outcomes in their SPM examinations.

1.1 Objectives

1.1.1 To develop a system that identifies individual students knowledges gaps and learning patterns through comprehensive performance tracking and data analysis

In the current educational landscape, students often struggle to identify their specific areas of weakness until examination results reveal the extent of their knowledge gaps that is too late for effective remediation. Qubo addresses this critical challenge by implementing a sophisticated analytics engine that continuously monitors student performance across all SPM subjects. The system tracks multiple dimensions of learning, including quiz scores, time spent on topics, question difficulty ratings, and study session patterns using Pomodoro-style tracking methodologies.

By analyzing this multifaceted data, the platform creates a comprehensive learning profile for each student, enabling precise identification of knowledge gaps at both macro (subject-level) and micro (topic-level) granularities. This objective ensures that students gain complete visibility into their strengths and weaknesses, allowing them to prioritize their study efforts effectively and address problem areas before they impact examination performance.

1.1.2 To implement AI-powered personalized learning recommendations that adapt to individual learning styles and optimize study efficiency through data-driven insights

Traditional study approaches often fail to accommodate the diverse learning styles and paces of individual students. Qubo employs advanced machine learning algorithms to generate personalized study recommendations tailored to each student's unique learning profile. The system analyzes performance data, learning velocity metrics, and behavioral patterns to provide customized guidance on optimal study time allocation, topic prioritization, and resource selection.

The platform categorizes students into visual, auditory, and kinesthetic learning styles, subsequently curating resources that align with their preferred learning modalities. By leveraging predictive analytics, Qubo forecasts examination performance based on current learning trajectories and provides early risk alerts for potential academic difficulties. This proactive approach enables students to make informed decisions about their study strategies and adjust their learning approaches to maximize efficiency and outcomes.

1.1.3 To create an automated quiz generation system using natural language processing that transforms textbook content into targeted practice assessments, enabling unlimited customized practice opportunities

Access to quality practice materials remains a significant barrier for many SPM students, particularly those from under-resourced backgrounds. Qubo revolutionizes practice assessment creation by implementing an NLP-powered quiz generation engine that automatically extracts key concepts from user-uploaded textbook pages and transforms them into targeted practice questions. This innovative feature eliminates the limitation of pre-made question banks and provides students with unlimited practice opportunities directly aligned with their study materials.

Students can simply upload images or PDFs of textbook pages, and the system intelligently generates multiple-choice questions, fill-in-the-blank exercises, and short-answer prompts based on the content. All generated quizzes are automatically saved to a personal library, enabling repeated practice and spaced repetition—a proven technique for long-term retention. This objective ensures that every student, regardless of their access to commercial study materials, can practice extensively and build confidence in their subject mastery.

1.2 Background

Malaysian secondary school students face significant challenges in managing their learning effectively across multiple subjects as they prepare for the Sijil Pelajaran Malaysia (SPM) examination. According to the 2024 SPM Results Analysis Report published by the Malaysian Examinations Board (Lembaga Peperiksaan Malaysia), while the overall pass rate remained stable at 97.5%, there are concerning trends in subject-specific performance that highlight the need for personalized learning interventions. For instance, Additional Mathematics recorded an alarming 24.9% failure rate with a GPMP of 5.62 despite having 19.9% of students achieving excellence grades. GPMP (Gred Purata Mata Pelajaran) means average subject grede point. Lower GPMP values (1.00 - 3.00) mean better overall performance (more students getting A+, A, A- grades) while higher GPMP values (4.00 - 7.00) mean weaker overall performance (more students getting lower grades like C, D, E, G). These statistics reveal a significant performance gap, suggesting that students struggle with identifying their knowledge gaps and allocating study time efficiently across different subjects.

The 2024 report further reveals critical disparities in academic achievement that underscore the inadequacy of one-size-fits-all teaching approaches. In core subjects like Bahasa Melayu and Bahasa Inggeris, the average grade points were 3.72 and 4.75 respectively, indicating room for improvement in language proficiency. More concerning is the performance in Science subjects: Physics recorded a failure rate of 0.4% with GPMP of 3.94, Chemistry showed 1.6% failure with GPMP of 4.52, and Biology had 0.8% failure with GPMP of 4.12. These varying performance levels across subjects demonstrate that students require differentiated support and personalized learning strategies to address their individual weaknesses effectively, particularly in subjects requiring higher-order analytical and problem-solving skills.

Furthermore, another research indicates that Malaysian students experience high academic stress, with 49.5% reporting moderate to high stress levels related to examinations (Ang & Huan, 2006). This stress is compounded by the traditional classroom approach that fails to accommodate diverse learning styles and paces (Ministry of Education Malaysia, 2013). Moreover, educators lack real-time visibility into individual student progress, limiting their ability to provide timely interventions. The 2024 analysis reveals that while urban schools generally perform better, rural students face additional challenges, with the urban-rural performance gap remaining significant across most subjects. For instance, in subjects like Ekonomi (GPMP 5.44) and Prinsip Perakaunan (GPMP 3.93), there is substantial variation in student performance, indicating that some students grasp concepts quickly while others struggle without adequate personalized support.

The COVID-19 pandemic has exacerbated these challenges, with the shift to online and hybrid learning exposing gaps in self-regulated learning skills (UNESCO, 2021). Current educational technology solutions in Malaysia are fragmented, focusing on either content delivery or assessment without integrating comprehensive analytics and personalized learning pathways (Ahmad et al., 2022). The 2024 SPM results demonstrate that students need more than just access to content, they require intelligent systems that can track their performance across multiple subjects, identify patterns in their learning behavior, and provide data-driven recommendations for improvement. With subjects ranging from core requirements to technical vocational courses showing varied performance outcomes, there is a clear need for an integrated platform that combines performance tracking, intelligent analytics, and personalized recommendations to support SPM students in their learning journey while providing educators with actionable insights for targeted interventions.

1.2.1 Comparision

Recent data from the Malaysian Ministry of Education reveals that only 31% of government schools have implemented any form of learning analytics system, and most of these are limited to basic attendance and grade tracking. Furthermore, SPM students have access to smartphones, only a few use educational apps regularly for exam preparation, citing lack of personalized content and engagement as primary barriers.

Table 1.1 below presents a comparative analysis of existing learning platforms available to Malaysian students, highlighting their strengths and limitations in addressing SPM preparation needs.

Table 1.1: Comparison of Existing Educational Platforms

Platform	Wayground (Old name: Quizizz)	Kahoot	Duolingo	Qubo (Proposed)
Analytics Dashboard	Basic	Limited	✔	✔
AI-Powered Quiz Generation	Pre-made only	Pre-made only	Adaptive	✔ Generated from user content
Predictive Analytics	✖	✖	✖	✔
Personalized Recommendations	Limited	✖	Algorithm-based	✔ Learning style-based
SPM-Specific Content	✖	✖	✖	✔
Gamification	✔	✔	✔	✔

This table 1.1 reveals that while existing platforms offer certain features, none provide the comprehensive, AI-driven, SPM-focused solution that Qubo delivers. The integration of advanced analytics, personalized recommendations, automated content generation, and predictive insights positions Qubo as a transformative tool for SPM preparation.

1.3 Target Market

1.3.1 Primary End User: SPM Students (Form 4 and Form 5)

The primary stakeholders are Malaysian secondary school students aged 16-17 preparing for their SPM examinations. With approximately 400,000 students taking the SPM annually, this represents a substantial user base with critical educational needs. These students will use the platform daily to complete AI-generated quizzes tailored to their current study materials, track their performance across all SPM subjects through comprehensive analytics dashboards, receive personalized study recommendations based on their learning patterns and performance data and access curated learning resources matched to their individual learning styles (visual, auditory, kinesthetic). Students benefit from enhanced self-awareness about their learning patterns, improved time management capabilities, and reduced examination anxiety through predictive insights and data-driven preparation strategies. The platform empowers them to take ownership of their learning journey and optimize their study efficiency.

1.3.2 Secondary End User: Educator and Teacher

Teachers and school administrators will utilize the platform to monitor class-wide performance trends and identify systemic knowledge gaps, identify at-risk students requiring immediate intervention through early warning alerts, access aggregated analytics to inform and refine teaching strategies, evaluate the effectiveness of different instructional approaches through performance metrics and provide targeted support and personalized interventions based on data-driven insights. Educators gain unprecedented visibility into learning outcomes beyond traditional assessments, enabling proactive rather than reactive teaching interventions. The platform reduces the administrative burden of manual progress tracking while providing sophisticated tools for differentiated instruction

1.3.3 Secondary End User: Parents and Guardians

As supportive stakeholders, parents can monitor their child's study habits, consistency, and overall progress, understand specific subject areas requiring additional support or resources, receive timely alerts about performance concerns or significant improvements, support effective goal-setting and maintain productive study routines at home. Parents gain transparent access to their child's academic progress, enabling them to provide appropriate support, encouragement, and resources to facilitate successful SPM preparation.

1.4 Problem Statement

1.4.1 Problems

1. Inability to Identify Knowledge Gaps and Optimize Study Strategies

Malaysian SPM students face a critical challenge in identifying their specific knowledge gaps across the broad SPM curriculum. Traditional learning approaches lack sophisticated feedback mechanisms that can pinpoint exactly where students struggle. Without real-time analytics and performance tracking, students often discover their weaknesses only after receiving examination results. Students struggle significantly with self-regulated learning, particularly in identifying areas requiring focused attention. (Chung, Noor, and Mathew , 2020)

The absence of data-driven insights forces students to rely on intuition rather than evidence when allocating study time, frequently resulting in inefficient preparation strategies. Students may spend excessive time on topics they have already mastered while neglecting areas of genuine weakness. This inefficiency is compounded by the lack of personalized study recommendations, leaving students without clear guidance on how to optimize their learning approach. The one-size-fits-all nature of traditional classroom instruction fails to accommodate individual learning paces and styles, further exacerbating the problem of ineffective study strategies.

2. High Academic Stress and Lack of Personalized Support

Malaysian students experience alarmingly high levels of academic stress, with Ang and Huan (2006) reporting that 49.5% of students suffer moderate to high stress levels related to examinations. This psychological burden stems partly from uncertainty about their academic readiness and the high-stakes nature of the SPM examination. The traditional classroom model, constrained by large student-teacher ratios and fixed curricula, cannot provide the individualized attention each student requires.

Students with diverse learning styles like visual, auditory, or kinesthetic are able to receive identical instruction and resources, regardless of their individual preferences and needs. This generic approach leaves many students disengaged and struggling to comprehend material presented in formats incompatible with their learning style. Furthermore, educators lack real-time visibility into individual student progress, limiting their ability to provide timely, targeted interventions. The absence of personalized support systems means that struggling students often fall further behind before their difficulties are recognized and addressed.

3. Limited Access to Quality Practice Materials and Adaptive Learning Resources

Access to comprehensive, high-quality practice materials remains a significant barrier for many SPM students, particularly those from under-resourced schools or economically disadvantaged backgrounds. Commercial test preparation resources can be prohibitively expensive, creating educational inequity. Even when practice materials are available, they are typically generic and not tailored to individual student needs or aligned with specific textbooks being used in different schools across Malaysia.

Current educational technology solutions in Malaysia are fragmented, focusing on either content delivery or assessment without integrating comprehensive analytics, adaptive learning, and personalized resource curation. Students lack tools that can automatically generate practice questions from their own textbooks or dynamically adjust difficulty based on performance. The COVID-19 pandemic has further exposed gaps in digital learning resources and students' self-regulated learning capabilities, with many struggling to effectively utilize available online materials.

1.4.2 Solutions

1. Intelligent Performance Tracking and Analytics System

Qubo implements a comprehensive performance tracking system that captures multidimensional learning data, including quiz scores, time spent on topics, question difficulty ratings, and study session patterns using Pomodoro-style tracking. This creates a complete learning profile for each student, enabling the platform to identify knowledge gaps with precision at both subject and topic levels.

The AI-powered analytics engine processes this data to provide interactive visualizations through dashboards showing subject-wise performance, learning velocity metrics, and comparisons with anonymized peer averages. Students gain immediate, actionable insights into their strengths and weaknesses, allowing them to make informed decisions about study time allocation and topic prioritization. The predictive analytics module forecasts examination performance based on current learning trajectories, provides early risk alerts for potential difficulties, and recommends optimal study strategies to maximize results.

2. Personalized Learning Recommendations and Adaptive Resource Curation

The platform employs advanced machine learning algorithms to generate personalized study recommendations tailored to each student's unique learning profile. By analyzing performance data, behavioral patterns, and learning styles, Qubo provides customized guidance on topic prioritization, study time allocation, and optimal learning strategies.

The intelligent resource recommendation engine categorizes students into visual, auditory, and kinesthetic learning styles, then curates learning resources (YouTube videos, articles, practice problems) that align with their preferred learning modalities and address specific performance gaps. The spaced repetition algorithm ensures that topics requiring reinforcement are presented at optimal intervals for long-term retention. The gamification framework incorporates streak tracking, achievement badges, and milestone celebrations to maintain engagement and build consistent study habits, reducing examination anxiety through structured, data-driven preparation.

3. NLP-Powered Automated Quiz Generation and Comprehensive SPM Coverage

Qubo revolutionizes practice assessment creation through its NLP-powered quiz generation engine that automatically extracts key concepts from user-uploaded textbook pages and transforms them into targeted practice questions. Students can upload images or PDFs of any textbook content, and the system intelligently generates multiple-choice questions, fill-in-the-blank exercises, and comprehension prompts aligned with SPM examination formats.

All generated quizzes are automatically saved to a personal library, enabling unlimited repeated practice and spaced repetition—a proven technique for long-term knowledge retention. This feature eliminates the limitation of pre-made question banks and ensures every student, regardless of economic background or school resources, has access to unlimited, high-quality practice materials directly aligned with their study content. The platform provides comprehensive coverage of all SPM subjects (Bahasa Melayu, English, Mathematics, Sciences, and electives), with subject-specific resource curation and culturally relevant content tailored to the Malaysian educational context.

1.5 Advantages and Contributions

1.5.1 Advantages

1. Integrated Analytics-First Approach

Unlike competitors such as Quizizz and Kahoot that focus primarily on gamified quizzing, or SchoolOnline Malaysia that emphasizes content delivery, Qubo prioritizes comprehensive learning analytics and actionable insights. Data serves as the core driver of personalized learning, with every feature designed to capture, analyze, and act upon student performance information. This analytics-first philosophy ensures that students receive evidence-based recommendations rather than generic study advice, fundamentally transforming how they approach SPM preparation.

2. SPM-Specific Optimization and Localization

Qubo is specifically tailored for the Malaysian education system and SPM curriculum, providing comprehensive coverage of all 17 SPM subjects with localized content recommendations and culturally relevant gamification elements. Unlike generic international platforms that lack awareness of Malaysian educational contexts, Qubo understands SPM examination formats, grading rubrics, and subject-specific requirements. This specialization ensures maximum relevance and effectiveness for Malaysian students, teachers, and parents.

3. Revolutionary AI-Powered Quiz Generation

The ability to transform any textbook page into targeted practice quizzes using natural language processing represents a significant technological advancement unavailable in competing platforms. This feature democratizes access to quality practice materials, ensuring that students from all socioeconomic backgrounds can generate unlimited, personalized assessments directly aligned with their study materials. The technology eliminates dependence on expensive commercial test preparation resources and pre-made question banks that may not align with specific textbook content.

4. Predictive Analytics and Early Intervention

Qubo's predictive analytics capabilities enable proactive rather than reactive educational interventions. By forecasting examination performance based on current learning trajectories and providing early risk alerts, the platform allows students, teachers, and parents to address potential difficulties before they become critical. This forward-looking approach, absent in competing platforms, transforms examination preparation from a reactive scramble into a strategic, data-informed process.

5. Learning Style Personalization

The explicit accommodation of visual, auditory, and kinesthetic learning preferences in resource recommendations creates genuinely personalized learning experiences. By matching content delivery formats to individual learning styles, Qubo enhances comprehension and retention more effectively than one-size-fits-all platforms. This personalization extends beyond content to include review frequency, and practice difficulty—all calibrated to individual student profiles.

6. Holistic Integration of Multiple Learning Dimensions

Qubo combines performance tracking, behavioral analytics, gamification, predictive insights, and automated content generation in a single, cohesive platform. This integration eliminates the need for students to juggle multiple disconnected tools, reducing cognitive load and improving efficiency. The platform's holistic approach ensures that all aspects of effective learning of assessment, feedback, practice, motivation, and planning work synergistically to optimize student outcomes.

1.5.2 Contributions

To Students:

1. Empowers self-directed learning through comprehensive, data-driven insights into personal strengths, weaknesses, and learning patterns
2. Significantly reduces examination anxiety through predictive performance tracking and evidence-based preparation strategies
3. Builds crucial metacognitive skills and learning awareness that extend beyond SPM to higher education and lifelong learning

4. Increases engagement and motivation through gamification elements that make studying rewarding and habit-forming
5. Provides equitable access to high-quality learning resources regardless of socioeconomic background or school location

To Educators:

1. Provides evidence-based decision-making tools that inform instructional strategies and resource allocation
2. Enables early intervention for struggling students through automated risk detection and performance alerts
3. Dramatically reduces the administrative burden of manual progress tracking and report generation
4. Facilitates differentiated instruction by providing detailed insights into individual student needs and learning profiles
5. Improves teaching effectiveness through granular performance analytics and feedback on instructional impact

To Community and Society:

1. Addresses educational equity by providing personalized, high-quality support regardless of school resources or geographic location
2. Contributes to improved SPM pass rates and academic outcomes at national level, supporting Malaysia's educational development goals
3. Aligns with Malaysia's aspiration toward data-driven education transformation as outlined in the Education Blueprint 2013-2025
4. Promotes lifelong learning habits through enhanced self-awareness and metacognitive skill development
5. Reduces educational stress and improves student mental health and well-being through structured, confidence-building preparation

1.6 Project Plan

1.6.1 Project Scope

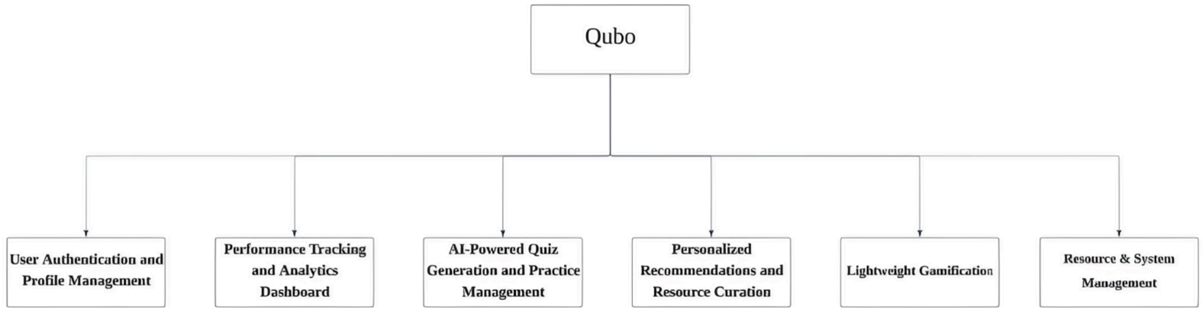

Figure 1.1: Modules of Qubo

The Qubo platform encompasses Three primary modules, each designed to address specific aspects of SPM preparation and learning optimization:

Module 1: User Authentication and Profile Management

This foundational module handles user registration, authentication, and role-based access control for students and educators. It manages secure storage of personal information, preferences, and account settings. Features include profile customization, learning style assessment, and privacy controls. The module ensures data security through encryption and implements OAuth-based authentication for seamless, secure access.

Module 2: Performance Tracking and Analytics Dashboard

The core analytics module captures comprehensive learning data including quiz performance, study session duration, topic difficulty ratings, and Pomodoro-style activity tracking. It processes this data through machine learning algorithms to identify knowledge gaps, learning velocity, and performance trends. The interactive dashboard visualizes subject-wise performance, progress over time, comparison with class averages, and predictive examination score forecasts. Real-time analytics provide immediate feedback on study effectiveness and learning patterns.

Module 3: AI-Powered Quiz Generation and Practice Management

This innovative module leverages natural language processing to automatically generate practice quizzes from user-uploaded textbook content. Students can photograph or upload PDF pages, and the NLP engine extracts key concepts, generates multiple-choice questions, and creates targeted assessments. The system implements spaced repetition algorithms to schedule review sessions optimally. All generated quizzes are stored in a personal library with performance tracking, enabling students to identify weak areas and practice repeatedly until mastery is achieved.

1.6.2 Milestone

Table 1.2: Project Milestones

ACTIVITIES	MILESTONE GOAL	START DATE	COMPLETION DATE
Approach and confirm project supervisor	Supervisor assigned and project scope confirmed	10/11/2025	19/11/2025
Prepare project proposal documentation	Complete proposal	20/11/2025	19/12/2025
Submit Individual Project Proposal	Approved proposal (Form 1, Form 2, Form 3)	10/11/2025	26/12/2025
Conduct literature review on AI/ML algorithms, learning analytics platforms, NLP for quiz generation, and EdTech solutions	Understanding of existing technologies and research gaps	27/12/2025	30/01/2026
Write Chapter 1: Introduction	Finalized introduction with problem statement, objectives, background, target market, and project scope	26/01/2026	06/02/2026
Research existing quiz generation systems and competitive landscape	Completed literature review covering AI/ML algorithms, learning analytics platforms, NLP for quiz generation, spaced repetition techniques, and existing EdTech solutions	09/02/2026	20/02/2026
Write Chapter 2: Research Background	Complete Chapter 2: Literature Review	09/02/2026	23/02/2026

Define system requirements and use cases	Defined Agile development process, technology stack (Python/TensorFlow, React, Supabase), hardware/software requirements, and detailed functional specifications for all modules	02/03/2026	05/03/2026
Develop project methodology	Clear development approach and analysis techniques defined	02/03/2026	10/03/2026
Write Chapter 3: Methodology and Requirements Analysis	Complete Chapter 3: Methodology and Requirements Analysis	02/03/2026	15/03/2026
Design system architecture and database schema	Finalized UI/UX wireframes, system architecture diagrams (API pipeline, database schema), API design specifications, and data flow documentation	16/03/2026	17/03/2026
Create UI/UX mockups and interface designs	Complete interface designs for all app screens	18/03/2026	27/03/2026
Write Chapter 4: System Design	Complete Chapter 4: System Design	16/03/2026	30/03/2026
Combine and format all chapters with cover page	Comprehensive compilation of Chapters 1-4 with proper formatting, references, and documentation	31/03/2026	13/04/2026
Setup development environment and project structure	Basic application framework ready	14/04/2026	20/04/2026
Develop quiz generation module	Functional quiz generation using AI/ML algorithms	21/04/2026	10/05/2026

Implement analytics dashboard	Working analytics dashboard displaying student performance metrics	11/05/2026	25/05/2026
Develop recommendation system	Recommendation system providing personalized learning paths	26/05/2026	05/06/2026
Integrate all modules into prototype	Functional prototype with core features (quiz generation, analytics dashboard) and documentation	06/06/2026	13/06/2026
Prepare test plan and test cases	Complete testing documentation and test cases for system validation	14/04/2026	18/06/2026
System preview with supervisor	Supervisor approval on system progress	14/04/2026	26/06/2026
Conduct final system testing	Fully integrated system with all modules (authentication, analytics, quiz generation, recommendations) verified and validated by supervisor	27/06/2026	10/07/2026
Submit draft FYP report	First draft documenting complete system implementation, testing results, analysis, and preliminary conclusions	11/07/2026	07/08/2026
Submission of Final FYP report and deliverables	Final report, complete source code, user documentation, deployment guide, and all digital deliverables submitted	08/08/2026	21/08/2026

1.6.3 Project Schedule Plan (Gantt Chart)

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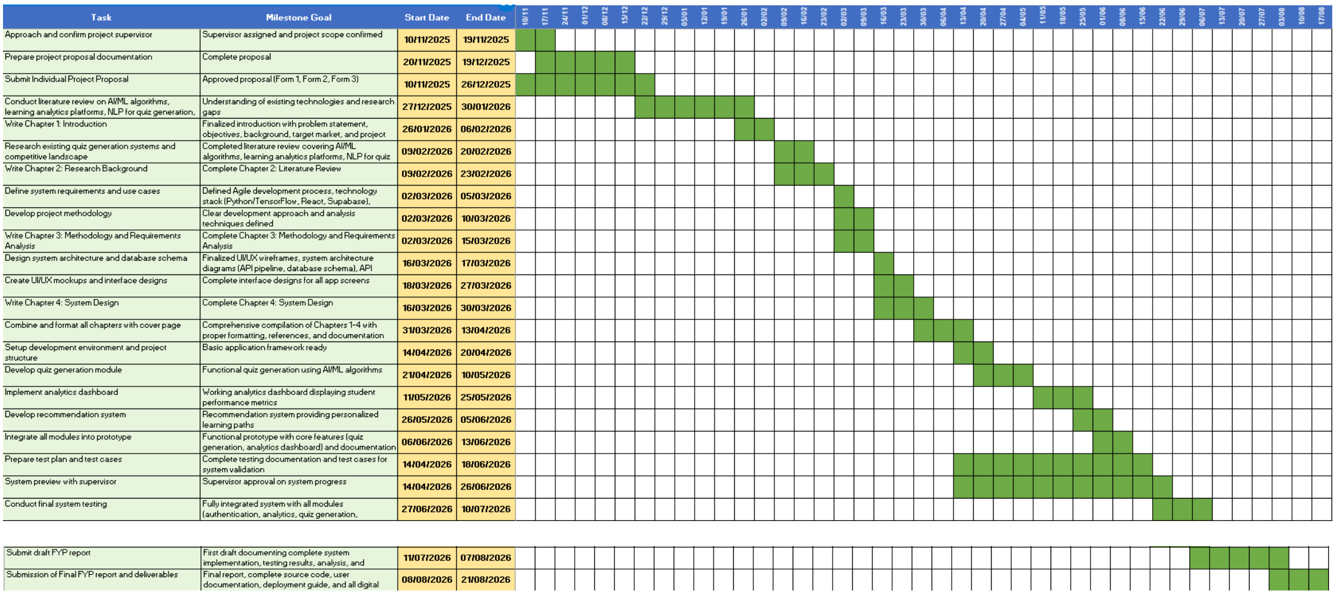

Task	Milestone Goal	Start Date	End Date
Approach and confirm project supervisor	Supervisor assigned and project scope confirmed	10/11/2025	19/11/2025
Prepare project proposal documentation	Complete proposal	20/11/2025	19/12/2025
Submit Individual Project Proposal	Approved proposal (Form 1, Form 2, Form 3)	10/11/2025	26/12/2025
Conduct literature review on AI/ML algorithms, learning analytics platforms, NLP for quiz generation,	Understanding of existing technologies and research gaps	27/12/2025	30/01/2026
Write Chapter 1: Introduction	Finalized introduction with problem statement, objectives, background, target market, and project	26/01/2026	06/02/2026
Research existing quiz generation systems and competitive landscape	Completed literature review covering AI/ML algorithms, learning analytics platforms, NLP for quiz	09/02/2026	20/02/2026
Write Chapter 2: Research Background	Complete Chapter 2: Literature Review	09/02/2026	23/02/2026
Define system requirements and use cases	Defined Agile development process, technology stack (Python/TensorFlow, React, Supabase),	02/03/2026	05/03/2026
Develop project methodology	Clear development approach and analysis techniques defined	02/03/2026	10/03/2026
Write Chapter 3: Methodology and Requirements Analysis	Complete Chapter 3: Methodology and Requirements Analysis	02/03/2026	15/03/2026
Design system architecture and database schema	Finalized UI/UX wireframes, system architecture diagrams (API pipeline, database schema), API	16/03/2026	17/03/2026
Create UI/UX mockups and interface designs	Complete interface designs for all app screens	18/03/2026	27/03/2026
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Setup development environment and project structure	Basic application framework ready	14/04/2026	20/04/2026
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Integrate all modules into prototype	Functional prototype with core features (quiz generation, analytics dashboard) and documentation	06/06/2026	13/06/2026
Prepare test plan and test cases	Complete testing documentation and test cases for system validation	14/04/2026	18/06/2026
System preview with supervisor	Supervisor approval on system progress	14/04/2026	26/06/2026
Conduct final system testing	Fully integrated system with all modules (authentication, analytics, quiz generation,	27/06/2026	10/07/2026
Submit draft FYP report	First draft documenting complete system implementation, testing results, analysis, and	11/07/2026	07/08/2026
Submission of Final FYP report and deliverables	Final report, complete source code, user documentation, deployment guide, and all digital	08/08/2026	21/08/2026

Figure 1.2: Gantt Chart of Qubo

1.7 Chapter Summary and Evaluation

This chapter has provided a comprehensive introduction to the Qubo AI-Powered Personalized Learning Platform, establishing the foundation for the project's development. The chapter began by contextualizing the challenges faced by Malaysian SPM students in an education system that increasingly demands personalized, data-driven approaches to learning.

The three primary objectives were articulated: developing an intelligent analytics system for identifying knowledge gaps, implementing AI-powered personalized recommendations, and creating an NLP-based automated quiz generation engine. Each objective addresses specific deficiencies in current educational technology offerings and directly responds to documented challenges in SPM preparation.

The background section established the educational context, highlighting research findings on student stress levels, self-regulated learning difficulties, and the impact of COVID-19 on educational delivery. The comparative analysis of existing platforms demonstrated that while various solutions offer individual features, none provides the comprehensive, integrated, SPM-specific approach that Qubo delivers.

The target market analysis identified primary users (Form 4 and Form 5 SPM students) and secondary stakeholders (educators and parents), detailing how each group will benefit from the platform's capabilities. The problem statement section systematically documented three critical challenges: inability to identify knowledge gaps, high academic stress with lack of personalized support, and limited access to quality practice materials. For each problem, corresponding solutions were proposed, demonstrating how Qubo's technological approach addresses these educational challenges.

The advantages and contributions section highlighted Qubo's distinctive features, its analytics-first approach, SPM-specific optimization, revolutionary quiz generation capability, predictive analytics, learning style personalization, and holistic integration that demonstrates clear differentiation from competing platforms. The contributions to students, educators, and society were articulated, emphasizing both immediate benefits and long-term educational impact.

Finally, the project plan section defined the scope through three primary modules (authentication, analytics,and quiz generation) and outlined eleven critical system functions. The milestone table provides a clear timeline for project execution, from proposal approval through final deliverable submission, ensuring accountability and systematic progress tracking.

This chapter successfully establishes the rationale, scope, and strategic direction for the Qubo project, providing a solid foundation for the detailed technical and methodological discussions that will follow in subsequent chapters. The comprehensive nature of this introduction ensures that all stakeholders (students, educators, supervisors, and evaluators) understand the project's significance, approach, and anticipated outcomes.

Chapter 2

Literature Review

2 Literature Review

2.1 Project Background

The Malaysian education system places significant emphasis on the Sijil Pelajaran Malaysia (SPM) examination, a high-stakes national assessment taken by Form 5 students that largely determines their academic and career trajectories. With over 400,000 candidates sitting for the SPM annually, the pressure surrounding this examination has given rise to widespread academic stress among secondary school students. A study by Ramli et al. (2018) published in the *Asian Journal of Psychiatry* found that approximately 49.5% of Malaysian secondary school students reported moderate to high levels of academic stress, with examination preparation cited as the primary stressor. This psychological burden is further compounded by students' limited ability to self-assess their own knowledge gaps, often resulting in inefficient and unstructured study habits.

Traditional classroom instruction in Malaysia operates predominantly within a one-size-fits-all framework, leaving little room for individualized learning. Hattie (2009), in his landmark meta-analysis *Visible Learning*, established that personalized, feedback-driven instruction produces among the highest effect sizes on student achievement, yet such approaches remain largely inaccessible in resource-constrained public school environments. This disparity is particularly acute in rural and semi-urban Malaysia, where students from lower-income households face additional barriers to quality supplementary educational resources. According to the World Bank, Malaysia's learning poverty rate stated that the proportion of children unable to read and understand a simple text by age 10 stood at approximately 48%, highlighting systemic gaps in foundational learning outcomes that persist into secondary education.

The rapid advancement of artificial intelligence and machine learning technologies has opened new possibilities for addressing these challenges through intelligent educational systems. Adaptive learning platforms powered by machine learning algorithms have demonstrated measurable improvements in student outcomes by dynamically tailoring content difficulty, pacing, and resource recommendations to individual learners. AI-driven personalization in education could improve learning efficiency by up to 20–40% compared to traditional instruction, underscoring the transformative potential of these technologies at scale.

Complementing adaptive learning, spaced repetition is a well-established cognitive science technique research on the forgetting curve, which demonstrates that information is retained more effectively when reviewed at systematically increasing intervals. Platforms such as Anki and Duolingo have successfully commercialized spaced repetition at scale; however, their content is largely generic and not aligned with the specific demands of the Malaysian SPM curriculum. This represents a critical gap in the current edtech landscape for Malaysian students.

Natural Language Processing (NLP) has similarly matured to a point where automated question generation from unstructured text is increasingly viable. Kurdi et al. (2020), in a systematic review of automatic question generation published in the *International Journal of Artificial Intelligence in Education*, found that NLP-based systems could generate educationally valid questions from textual sources with increasing accuracy, reducing the time educators spend on assessment creation and enabling scalable personalized practice. Despite this progress, no widely adopted platform currently enables Malaysian SPM students to generate curriculum-aligned practice questions directly from their own textbooks.

It is within this confluence of factors such as high examination pressure, inequitable access to personalized learning support, and the maturation of AI-driven educational technologies that Qubo is conceptualized. Qubo aims to bridge the gap between the advanced capabilities of modern edtech and the specific, underserved needs of Malaysian SPM students by integrating intelligent analytics, personalized recommendations, NLP-powered quiz generation, and lightweight gamification into a single, cohesive platform.

2.1.1 SPM Education Landscape and Student Challenges

Malaysian SPM examination system, enrollment statistics, subject requirements, and the documented academic pressures students face. Covers research on self-regulated learning deficiencies among Malaysian secondary students and how the lack of personalized feedback contributes to inefficient study habits and under performance.

2.2 Literature Review

2.2.1 Machine Learning and Learning Analytics

Machine learning (ML) is a subset of artificial intelligence that enables systems to learn from data and improve their performance over time without being explicitly programmed. In educational contexts, ML has been widely applied to analyze student performance data, identify learning patterns, and generate actionable insights through learning analytics dashboards.

One of the most influential ML models in education is **Bayesian Knowledge Tracing (BKT)**, introduced by Corbett and Anderson (1994), which models a student's knowledge state as a hidden variable that updates dynamically based on their quiz responses. This allows the system to estimate the probability that a student has mastered a particular concept at any given point in time. Building on this, **Deep Knowledge Tracing (DKT)**, proposed by Piech et al. (2015) from Stanford University, applied Long Short-Term Memory (LSTM) recurrent neural networks to knowledge tracing and achieved significantly higher prediction accuracy than BKT across large-scale datasets from Khan Academy and ASSISTments.

In practice, platforms like **Knewton** (now integrated into Wiley) and **Carnegie Learning's MATHia** use ML-driven adaptive engines to continuously adjust content difficulty and pacing based on individual student performance. Carnegie Learning reported in a 2019 study that students using MATHia outperformed control groups by 23 percentile points in algebra assessments, demonstrating measurable ML-driven learning gains (Carnegie Learning, 2019).

Limitation: A key challenge is that ML models require large volumes of training data to perform accurately. For a new platform like Qubo with a limited initial user base, cold start problems may reduce early prediction accuracy.

How Qubo addresses this: Qubo mitigates this by incorporating rule-based heuristics during the early stages of a student's usage, gradually transitioning to ML-driven recommendations as sufficient performance data is accumulated.

2.2.2 Spaced Repetition Algorithm

Spaced repetition is a learning technique grounded in cognitive psychology, specifically in Ebbinghaus's (1885) research on the forgetting curve, which demonstrated that memory retention decays exponentially over time unless information is reviewed at strategically spaced intervals. Ebbinghaus found that approximately 56% of newly learned information is forgotten within one hour, and up to 66% within one day without reinforcement, establishing the scientific basis for systematic review scheduling.

The most widely implemented spaced repetition algorithm is SM-2, developed by Piotr Wozniak in 1987 for the SuperMemo software. The SM-2 algorithm calculates the optimal interval between reviews based on the student's self-reported difficulty rating after each card review, using the formula:

$$I(n) = I(n-1) \times EF$$

where I is the inter-repetition interval in days and EF (ease factor) is a difficulty modifier that adjusts based on student performance. This algorithm forms the backbone of Anki, one of the most widely used spaced repetition platforms globally, with over 10 million active users (Anki, 2023).

Research by Cepeda et al. (2008), published in *Psychological Science*, conducted a large-scale study involving over 1,350 participants and found that spaced practice produced retention rates up to 64% higher than massed practice (cramming) across a one-month retention interval. This is particularly relevant for SPM students who commonly resort to last-minute intensive cramming before examinations.

Limitation: Existing platforms like Anki are content-agnostic and require students to manually create their own flashcard decks, which is time-consuming and does not guarantee alignment with the SPM syllabus.

How Qubo addresses this: Qubo automates flashcard and quiz generation directly from SPM textbook content using NLP, and schedules reviews using spaced repetition intervals, removing the manual burden from students while ensuring curriculum alignment.

2.2.3 Natural Language Processing for Automated Quiz Generation

Natural Language Processing (NLP) is a branch of artificial intelligence concerned with enabling computers to understand, interpret, and generate human language (Jurafsky & Martin, 2023). In educational technology, one of the most impactful NLP applications is Automatic Question Generation (AQG), which involves generating meaningful and educationally valid questions from unstructured textual content such as textbooks or lecture notes.

Kurdi et al. (2020) conducted a systematic review of 93 studies on AQG published in the *International Journal of Artificial Intelligence in Education*, concluding that transformer-based models significantly outperform earlier rule-based and template-based approaches in generating grammatically correct and contextually relevant questions. Specifically, the T5 (Text-to-Text Transfer Transformer) model developed by Google (Raffel et al., 2020) has demonstrated strong performance on question generation benchmarks, treating the task as a text-to-text problem where the input is a passage and the output is a generated question-answer pair.

A real-world example is Quizbot, a platform developed by researchers at the National University of Singapore, which uses NLP to generate multiple-choice questions from uploaded lecture slides and has been tested with over 500 university students, reporting 78% question acceptability from educator evaluators (Gao et al., 2018).

Another relevant example is Khanmigo, Khan Academy's AI-powered tutor built on GPT-4, which demonstrates that large language models can generate contextually appropriate educational questions from curriculum content at scale (Khan Academy, 2023).

Limitation: A key challenge in AQG is distractor quality — the incorrect answer choices in multiple-choice questions are often too obviously wrong, reducing the diagnostic value of the quiz. Additionally, OCR accuracy when processing textbook images can introduce errors into the input text, degrading question quality.

How Qubo addresses this**:** Qubo implements a preprocessing pipeline that cleans OCR output before passing it to the NLP model, and uses semantic similarity measures to generate plausible distractors that are meaningfully related to the correct answer but distinctly incorrect.

2.2.4 Predictive Analytics in Education

Predictive analytics involves applying statistical algorithms and machine learning techniques to historical data in order to forecast future outcomes (Siemens & Long, 2011). In education, predictive analytics has been extensively studied within the field of Learning Analytics, defined by the Society for Learning Analytics Research (SOLAR) as "the measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs."

One of the most well-documented applications is early warning systems (EWS) within Learning Management Systems (LMS). Macfadyen and Dawson (2010) analyzed data from 118 students using Blackboard LMS and identified that just three behavioral variables — total number of discussion posts, messages sent, and assessments completed — could predict final grades with 81% accuracy. This demonstrated that even relatively simple behavioral signals carry strong predictive power for academic outcomes.

The Open University Learning Analytics Dataset (OULAD), one of the largest publicly available educational datasets, contains data from over 32,000 students across 22 courses. Multiple studies using OULAD have achieved over 80% accuracy in predicting student dropout and final grade classification using gradient boosting and random forest algorithms (Hlosta et al., 2017).

In the Malaysian context, Affendi et al. (2021) applied predictive analytics to secondary school student data and found that attendance patterns, assignment submission rates, and formative assessment scores were the strongest predictors of SPM performance, with their model achieving 76% classification accuracy.

Limitation: Predictive models can produce false positives, incorrectly flagging students as at-risk, which may cause unnecessary anxiety. Additionally, models trained on historical data may not generalize well to new student cohorts with different behavioral patterns.

How Qubo addresses this: Qubo presents predictions as performance trend indicators rather than definitive risk labels, framing insights positively as areas for improvement rather than warnings of failure, reducing potential negative psychological impact.

2.3 Feasibility Study

Technical Feasibility

State that Qubo will be developed using React (frontend), Python with TensorFlow (AI/ML backend), and Supabase (database). Justify each choice: React is widely adopted and well-documented, TensorFlow has proven ML model support, Supabase offers real-time data sync at low cost. Mention that NLP libraries such as HuggingFace Transformers are open-source and well-validated in research. Conclude that the technology stack is technically feasible.

Economic Feasibility

All core technologies (Python, TensorFlow, React, Supabase free tier, HuggingFace) are open-source or freemium. Development hardware is already owned. No licensing costs are required. Therefore the project is economically feasible with minimal overhead.

Operational Feasibility

Qubo only requires a smartphone or computer with internet access and a browser. Students can upload textbook images using their phone camera. The platform is designed to integrate naturally into existing study routines rather than replacing them. Therefore it is operationally feasible for the target user group.

2.4 Chapter Summary and Evaluation

This chapter provided a comprehensive review of the literature and foundational concepts underpinning the development of Qubo. The project background established the academic challenges faced by Malaysian SPM students, including high examination stress, limited access to personalized learning support, and the inability to accurately self-identify knowledge gaps. The literature review examined four core technologies: machine learning, spaced repetition, natural language processing and predictive analytics, each supported by established academic research demonstrating their effectiveness in educational contexts. A comparative analysis of existing platforms further revealed that no current solution adequately addresses the specific needs of SPM students in terms of curriculum alignment, intelligent analytics, and automated content generation. The feasibility study confirmed that Qubo is technically, economically, and operationally viable given the open-source nature of its technology stack and the accessibility of modern mobile devices. These findings collectively establish the theoretical and practical foundation for the system design and development decisions presented in the subsequent chapters.

Chapter 3

Methodology and Requirements Analysis

3 Methodology and Requirements Analysis

This chapter presents the methodology and requirements analysis for Qubo, an AI-powered personalized learning platform designed for Malaysian SPM students. It outlines the software development model chosen for the project and explains how it is applied across the three core modules of the platform. The chapter also discusses the technical approaches employed to implement the platform's AI-powered capabilities, including the Gemini Flash API-powered quiz generation pipeline for automated quiz generation and the machine learning-based analytics engine. Following this, the chapter presents a comprehensive requirements analysis covering both functional and non-functional requirements across all modules. Finally, the development environment and tools used throughout the project are summarized.

3.1 Methodology

3.1.1 Software Development Model

The Agile Software Development Methodology is chosen to develop Qubo, an AI-Powered Personalized Learning Platform. Agile is selected because the platform is designed around iterative delivery and continuous feedback, both of which are essential for building and validating AI-driven features that depend on evolving user requirements and model performance. Unlike a rigid sequential approach, Agile allows each module to be developed, tested, and refined in short sprints before it is integrated with the rest of the platform, ensuring that the system grows in a controlled and responsive manner.

The platform is decomposed into three primary modules, each developed as a distinct increment. These modules are developed in the following order:

1.User Authentication and Profile Management Module'

2.Performance Tracking and Analytics Dashboard Module

3.AI-Powered Quiz Generation and Practice Management Module

Each module goes through four Agile phases: Requirements Analysis, Design, Implementation, and Testing, before it is integrated with all previously completed modules. For example, once the User Authentication and Profile Management Module is completed and tested, it is integrated as the foundation for the next sprint. When the Performance Tracking and Analytics Dashboard Module is ready, it is merged with the authentication module. The Quiz Generation Module is then integrated with both, and so on. This incremental integration ensures that each new layer of functionality is verified to work correctly with everything already built.

Once all three modules are completed and individually tested, they are brought together and the full platform undergoes end-to-end integration testing. This final testing phase checks that all modules communicate and function correctly as one complete system. Any cross-module issues discovered during this phase are identified and resolved before the project is submitted.

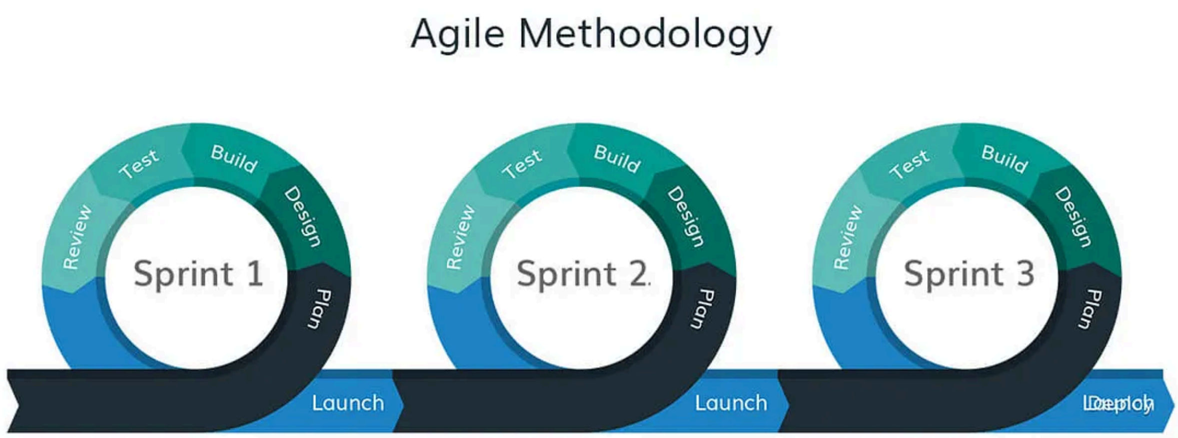


Figure 3.1: Agile for Qubo Development

3.1.2 AI Quiz Generation Using Gemini Flash API

The core innovation of Qubo is an automated quiz generation engine powered by Google Gemini Flash API (gemini-2.0-flash), a multimodal large language model that forms the backbone of the AI-Powered Quiz Generation and Practice Management Module. When a student uploads a textbook page as an image or PDF, the system sends the content directly to the Gemini Flash API along with a structured prompt instructing the model to generate SPM-format practice questions. Since Gemini Flash has built-in vision capability, no separate OCR step is required, the model reads the image directly and understands both printed text and diagrams.

The API is called with a structured prompt specifying the question types required (multiple-choice with 4 options, fill-in-the-blank, and short-answer), the target format (JSON), and the educational context (SPM examination). Gemini Flash returns a structured JSON response containing the generated questions, answer choices, and correct answers, which the backend parses, validates, and stores in Supabase.

3.1.3 Pre-processing

Before textbook content is fed into the NLP model, it passes through a pre-processing pipeline designed to standardize and clean the input. Raw images and PDF scans can vary significantly in quality, orientation, and formatting, and feeding them directly into the model without preparation would reduce the accuracy of the generated questions.

The pre-processing steps applied are:

1. **Image and document preparation** — resize/compress images before sending to the API to reduce payload size and API cost. PDFs are split into individual pages and converted to JPEG before upload.

2. **Prompt construction** — a structured prompt is built dynamically, injecting the subject context (e.g. "Physics SPM") and question format requirements. The prompt instructs Gemini to respond strictly in JSON format.

3.1.4 Post-processing

After the Gemini Flash API returns a response, a post-processing stage is performed before the generated questions are presented to the student or stored in the quiz library.

The API is prompted to return its output strictly in JSON format, where each question object contains four fields: the question text, an array of four answer options, the correct answer key, and the question type (multiple-choice, fill-in-the-blank, or short-answer). The first post-processing step is JSON schema validation, which checks that every question object returned by the API contains all required fields and that none of the fields are empty or malformed. Any question object that fails this validation is discarded automatically without being shown to the student.

Following schema validation, a question completeness check is applied. This check verifies that the question text is meaningful and not a repetition of another question in the same batch, that each multiple-choice question contains four distinct answer options with no duplicates, and that the correct answer key corresponds to one of the provided options. Questions that fail the completeness check are also discarded.

A minimum threshold of three valid questions per upload is enforced. If the number of questions that pass both the schema validation and completeness check falls below three, the system notifies the student that the uploaded content did not produce enough valid questions and prompts them to upload a clearer or more content-rich page.

Once questions pass all post-processing checks, they are tagged with metadata including the subject, topic, and question type before being stored in the student's personal quiz library in Supabase. The spaced repetition scheduler then assigns each accepted question an initial review date based on the SM-2 algorithm, which determines when the question should next appear in the student's practice queue to maximize long-term retention.

3.1.5 Machine Learning-Based Analytics Engine

The Performance Tracking and Analytics Dashboard Module relies on a machine learning-based analytics engine to process raw learning data into actionable insights. As students complete quizzes, log study sessions, and interact with the platform, data points including quiz scores, time-on-task, question difficulty ratings, error patterns, and session frequency are continuously captured and stored in Supabase.

The analytics engine applies supervised and unsupervised learning techniques to this data. A classification model identifies each student's learning style (visual, auditory, or kinesthetic) based on their interaction patterns and self-assessment responses collected during onboarding. A regression model trained on historical performance data predicts future examination scores based on current learning trajectories, providing early risk alerts when a student's projected grade falls below their target. Clustering algorithms group students by performance profiles, enabling comparison with anonymized peer averages and identification of common knowledge gap patterns across a class.

The analytics engine recalculates predictions each time new quiz or session data is submitted by the student within the platform, improving its accuracy over time. All model inferences are performed server-side through a Python backend API using TensorFlow for lightweight ML inference ensuring that the mobile frontend remains lightweight and responsive.

3.1.6 Integration and Evaluation of the AI Pipeline

The Gemini Flash API quiz generation pipeline and the ML analytics engine are integrated into a unified AI backend deployed as a RESTful API. The React-based web frontend and mobile interface communicate with this backend to submit images to the Gemini Flash API, retrieve generated questions, receive analytics insights, and fetch personalized study recommendations. The Supabase database serves as the central data store for user profiles, quiz libraries, performance records, and session history.

The complete pipeline is evaluated on two key metrics: quiz generation accuracy (measured by the percentage of questions rated as relevant and well-formed by test users) and analytics prediction accuracy (measured by the mean absolute error between predicted and actual quiz performance). These metrics are tracked during development and testing to ensure that the AI components meet the quality thresholds defined in the non-functional requirements.

3.2 Requirement Analysis

3.2.1 Functional Requirement

Table 3.1: Functional Requirements

Module	Functional Requirement
1.0 User Authentication and Profile Management	1.1 The system shall allow user to register an account
	1.2 The system shall allow the user to reset their password.
	1.3 The system shall allow the user to login to his/her respective account.
	1.4 The system shall allow the user to logout his/her account.
	1.5 The system shall allow the user to update their profile information (username, profile picture, and target subjects)
	1.6 The system shall support role-based access control for student and educator roles.
	1.7 The system shall allow students to input personal academic details (form level, school, target SPM grade).
	1.8 The system shall allow students to select the SPM subjects they are studying.
	1.9 The system shall allow students to set a target examination date and study goal
	1.10 The system shall temporarily lock the account after 5 consecutive failed login attempts.
Performance Tracking and Analytics Dashboard	2.1 The system shall record study sessions with subject, topic, duration, and Pomodoro cycles completed
	2.2 The system shall record quiz attempt results including score, time taken, and per-question correctness.
	2.3 The system shall provide an interactive dashboard displaying learning analytics visualizations including bar charts, radar charts, and trend lines.

	2.4 The system shall identify and highlight the student's weakest topics based on quiz performance data.
	2.5 The system shall generate predictive exam score forecasts based on current learning trajectories, updated after each quiz attempt.
	2.6 The system shall issue early-warning notifications to students whose predicted exam scores fall below a configurable threshold (default: 50%).
	2.7 The system shall display a learning velocity metric showing rate of topic mastery improvement over time
	2.8 The system shall allow educators to view class-wide analytics and individual student performance summaries.
	2.9 The system shall allow users to filter analytics by subject, date range, or topic.
	2.10 The system shall generate and export a student progress report in PDF format.
AI-Powered Quiz Generation and Practice Management	3.1 The system shall allow students to upload textbook page images (JPEG/PNG) or PDF for quiz generation.
	3.2 The system shall send the uploaded content to the Gemini Flash API to extract key concepts and generate practice questions.
	3.3 The system shall generate multiple-choice questions with plausible distractors from the extracted content.
	3.4 The system shall generate fill-in-the-blank and short-answer questions from the extracted content.
	3.5 The system shall display the generated quiz for the student to attempt immediately or save for later.
	3.6 The system shall automatically save all generated quizzes to the student's personal quiz library.
	3.7 The system shall display instant feedback and correct answers after each quiz attempt.

	3.8 The system shall implement a spaced repetition algorithm to schedule topic review intervals based on recall performance.
	3.9 The system shall allow students to search and filter their quiz library by subject or topic.
	3.10 The system shall allow students to delete quizzes from their personal library.

3.2.2 Non-Functional Requirements

Table 3.2: Non-Functional Requirement

Non-Functional Requirements	
1.0 Reliability	1.1 The system shall achieve a minimum system uptime of 98% availability to ensure consistent access for students and educators at all times.
	1.2 The system shall display a clear and informative error message instead of crashing or freezing when an unexpected error occurs.
	1.3 The system shall ensure that no user data, quiz records, or study progress is lost in the event of an unexpected shutdown or network failure.
	1.4 The Gemini Flash API quiz generation pipeline shall return a result or an appropriate error notification within 30 seconds of a student uploading content.
2.0 Performance	2.1 Every dashboard screen and page shall fully load within 5 seconds under normal network conditions.
	2.2 The quiz generation pipeline shall produce at least three questions within 30 seconds of a content upload.
	2.3 Analytics charts and visualizations shall finish rendering within 10 seconds after the user navigates to the dashboard.
	2.4 The predicted examination score and risk alerts shall update within 10 seconds after a new quiz attempt is completed.
	2.5 The Gemini Flash API response shall achieve a minimum valid question acceptance rate of 75% based on JSON schema validation and question completeness checks.
	2.6 The predictive analytics model shall achieve a mean absolute error (MAE) of no greater than 10 percentage points when predicting quiz performance

3.0 Security	3.1 User authentication shall enforce a maximum of five (5) failed login attempts
	3.2 Each user's personal data and quiz library stored in Supabase shall only be accessible to that specific user.
4.0 Usability	4.1 The platform shall be usable by students without any technical experience, requiring no onboarding tutorial to navigate core features.
	4.2 The platform shall support both Bahasa Melayu and English language interfaces.
	4.3 The platform shall display a clear confirmation message after every significant user action such as saving a quiz, completing a study session, or updating a profile.
5.0 Maintainability	5.1 The codebase shall follow a clean architecture pattern separating frontend, backend API, and AI pipeline layers.
	5.2 Each of the three modules shall be independently deployable so that one module can be updated or patched without affecting the others.

3.2.3 Development Environment

Table 3.3: Development Environment

Resources	Tools
Hardware	Laptop Specification: ● I7-10750H CPU ● NVIDIA GeForce GTX 1650 with Max-Q Design ● 16 GB RAM ● 512 GB SSD Peripheral ● Mouse ● Keyboard
Development IDE	● Visual Studio Code
Programming Language	● React (frontend) ● Tailwind CSS (CSS framework) ● Python with TensorFlow (AI/ML backend)
Database	● Supabase
API	● Gemini Flash API
Version Control	● Git / GitHub
Operating System	● Windows 11

3.3 Chapter Summary and Evaluation

This chapter presented the methodology and system requirements for the development of the Qubo AI-Powered Personalized Learning Platform. It began by introducing the Agile Software Development Methodology as the chosen development approach, explaining how each of the three modules is developed, tested, and integrated in iterative sprints to ensure that the platform grows in a structured, feedback-driven, and manageable manner.

The AI pipeline sections described the two core technical components of the platform. The Gemini Flash API-powered quiz generation pipeline was outlined in detail, covering image pre-processing before API submission, the Gemini Flash API call for question generation, and covering image pre-processing before API submission, the Gemini Flash API call for question generation, and the post-processing JSON validation mechanism that ensures only well-formed questions reach the student's quiz library. The machine learning analytics engine was also described, including how it processes raw performance data to identify learning patterns, predict examination scores, and generate early risk alerts.

The functional requirements section defined the expected behaviour of all three modules, covering user authentication and profile management, performance tracking and analytics and AI-powered quiz generation and practice management. Thirty functional requirements in total were specified across the three modules, ensuring comprehensive coverage of all user interactions and system behaviours. The non-functional requirements established quality standards across five dimensions: reliability, performance, security, usability, and maintainability.

Finally, the development environment section outlined the Hardware, Development IDE, Programming Language, Database, API, Version Control, Operating System and supporting tools used throughout the project. Overall, this chapter establishes a clear and technically grounded methodology that serves as a solid foundation for the system design and implementation work in the subsequent chapters.

Chapter 4

System Design

4 System Design

4.1 User Interface Design

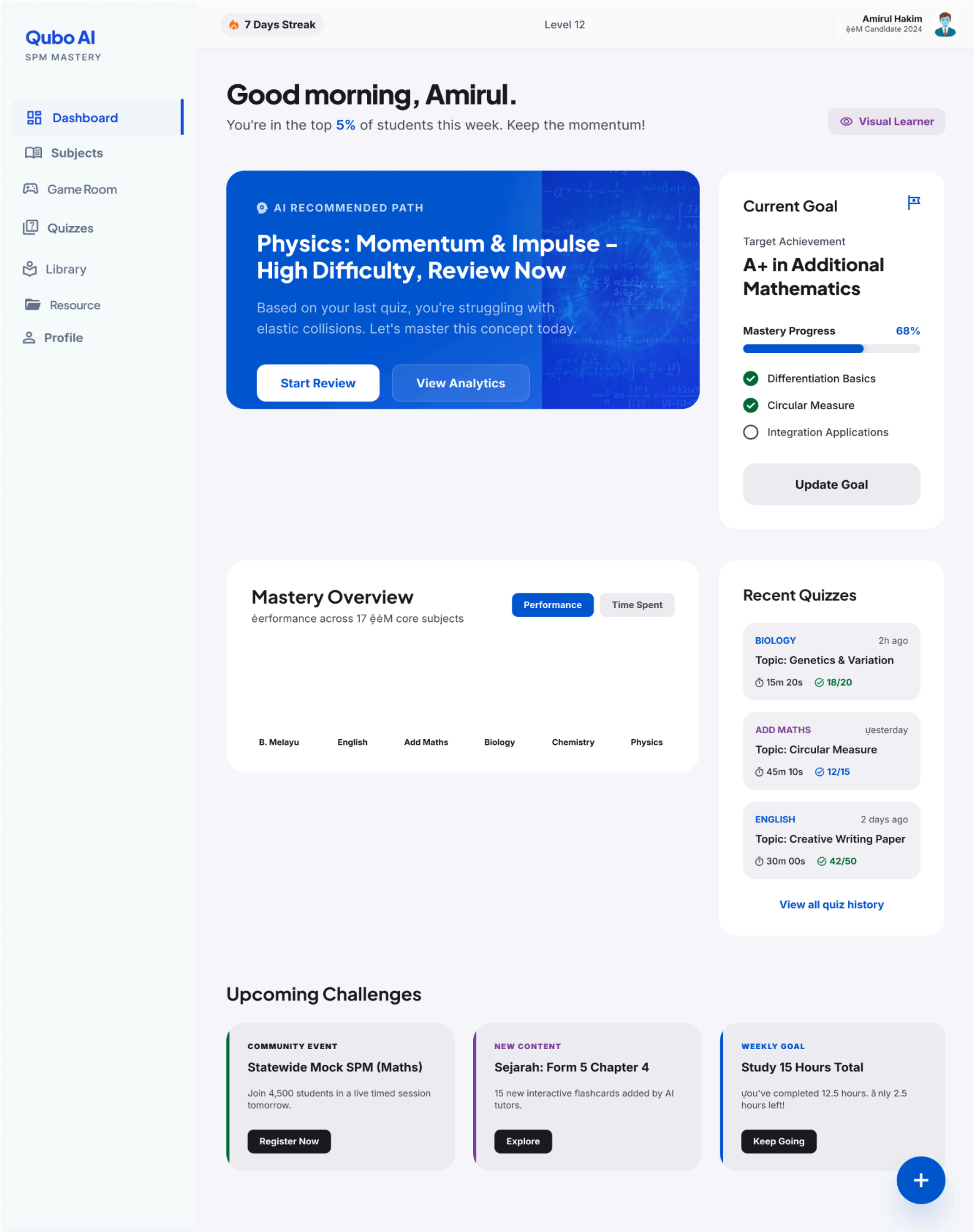


Figure 4.1: Main Page (Student Dashboard)

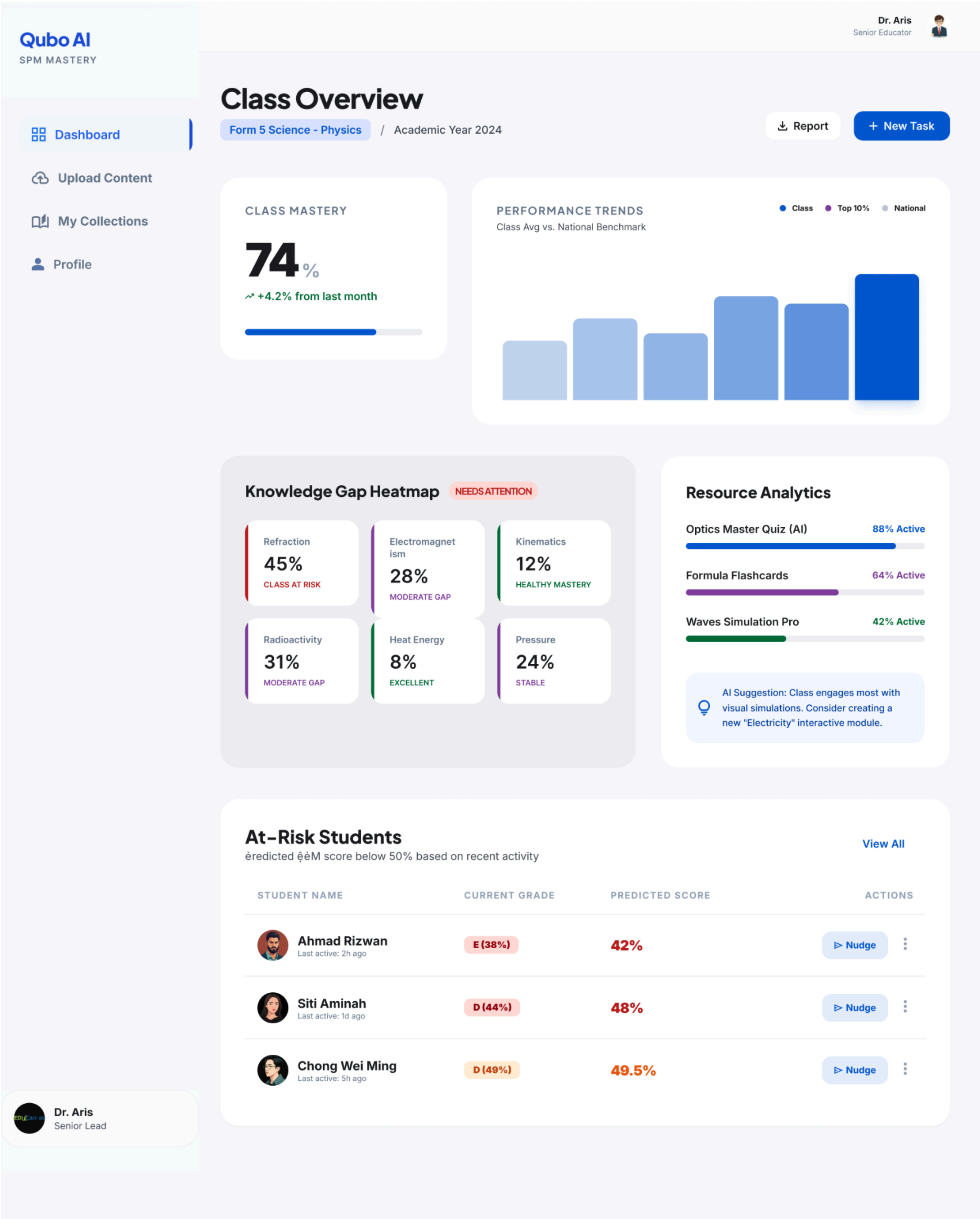


Figure 4.2: Main Page (Educator Dashboard)

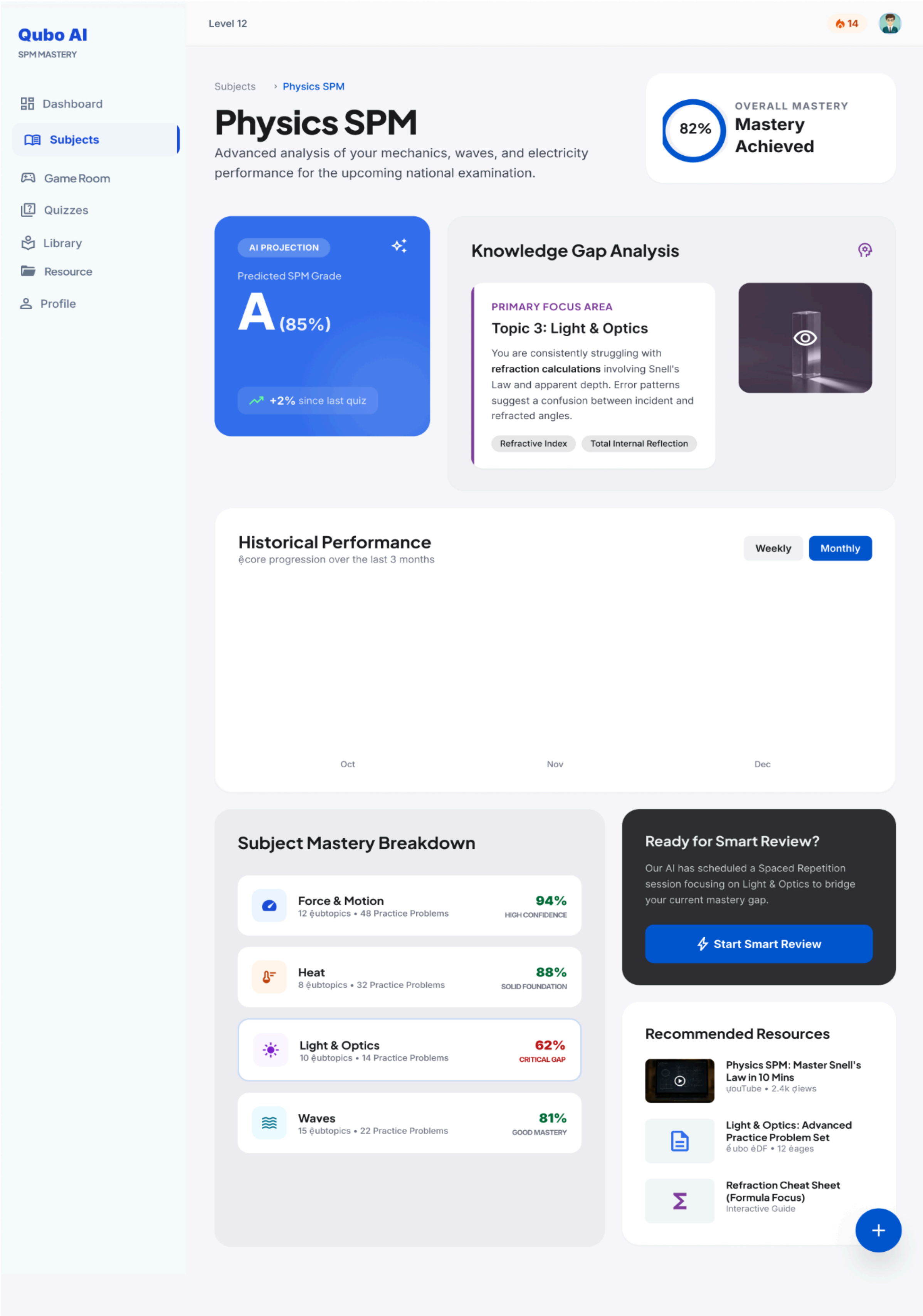


Figure 4.3: Performance Tracking and Analytics Page

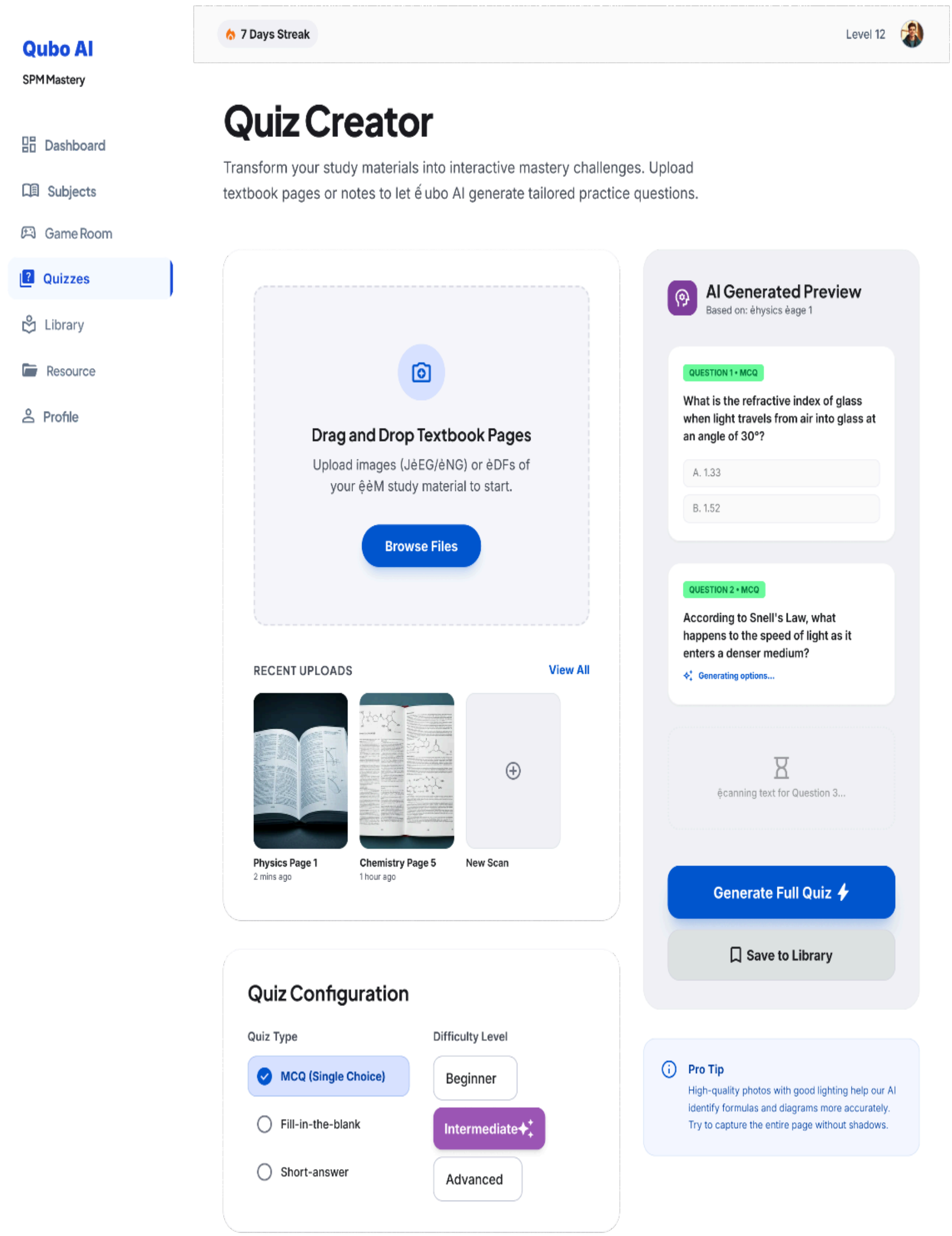


Figure 4.4: AI Quiz Generation Page

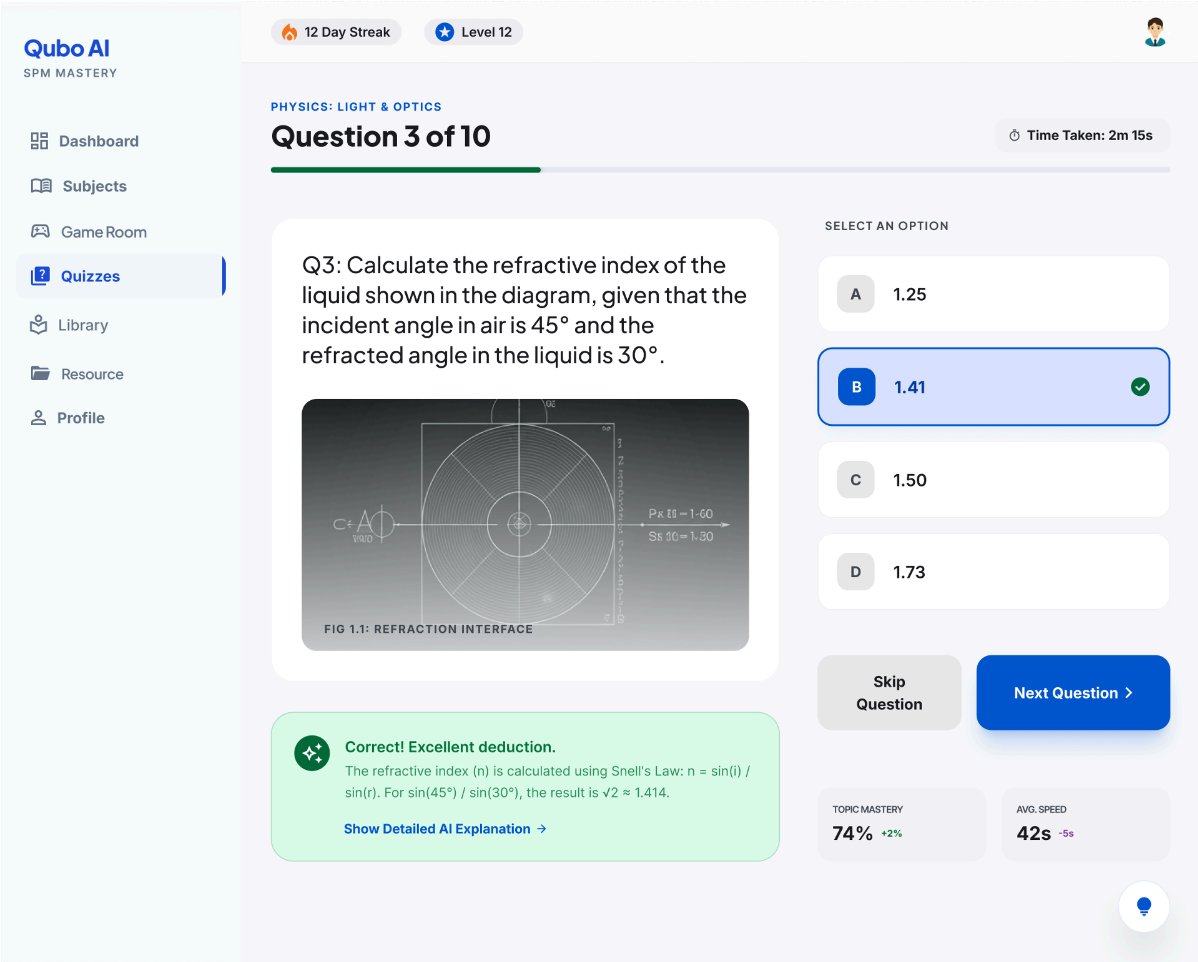


Figure 4.5: Quiz Experience

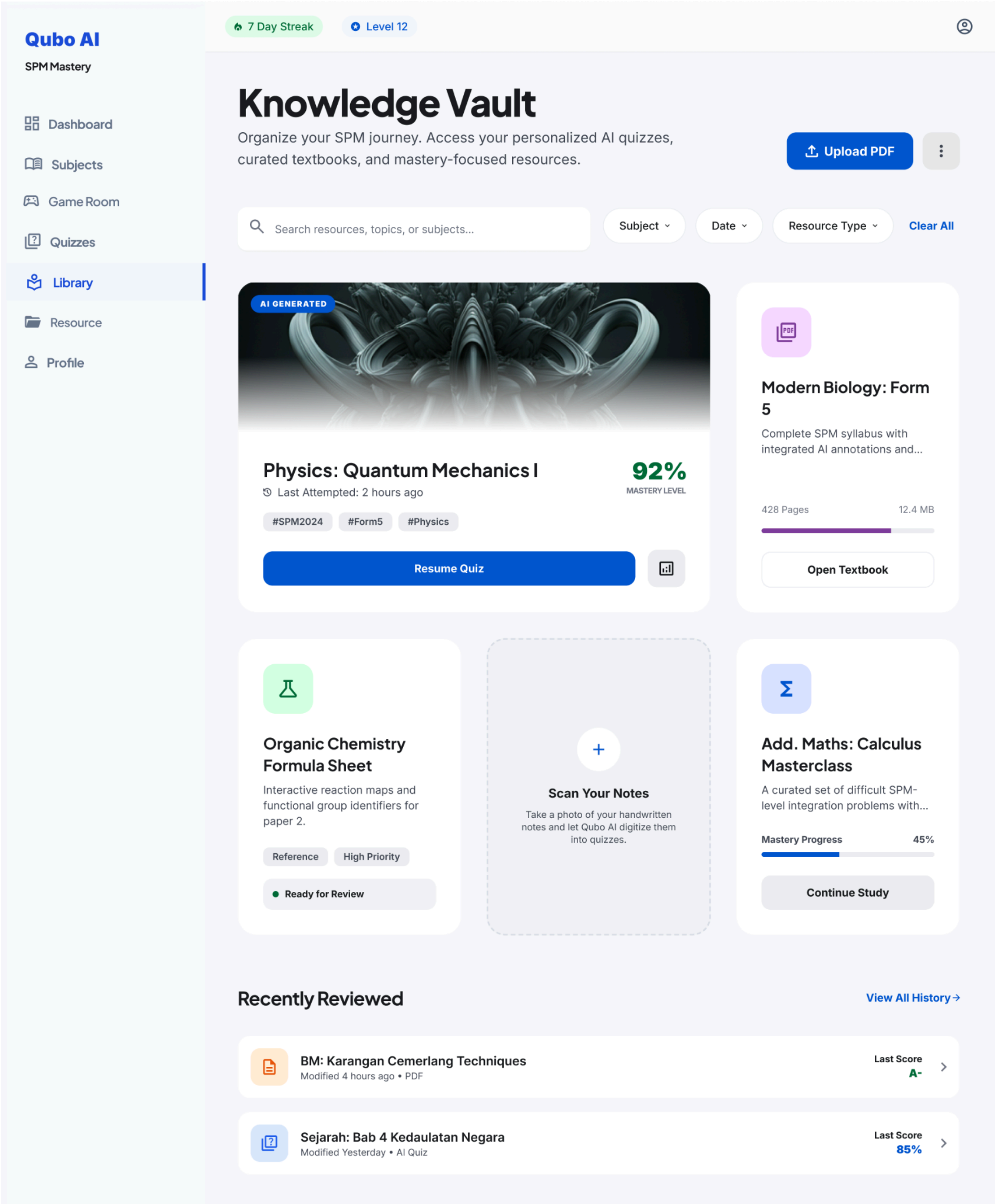


Figure 4.6: Library Page

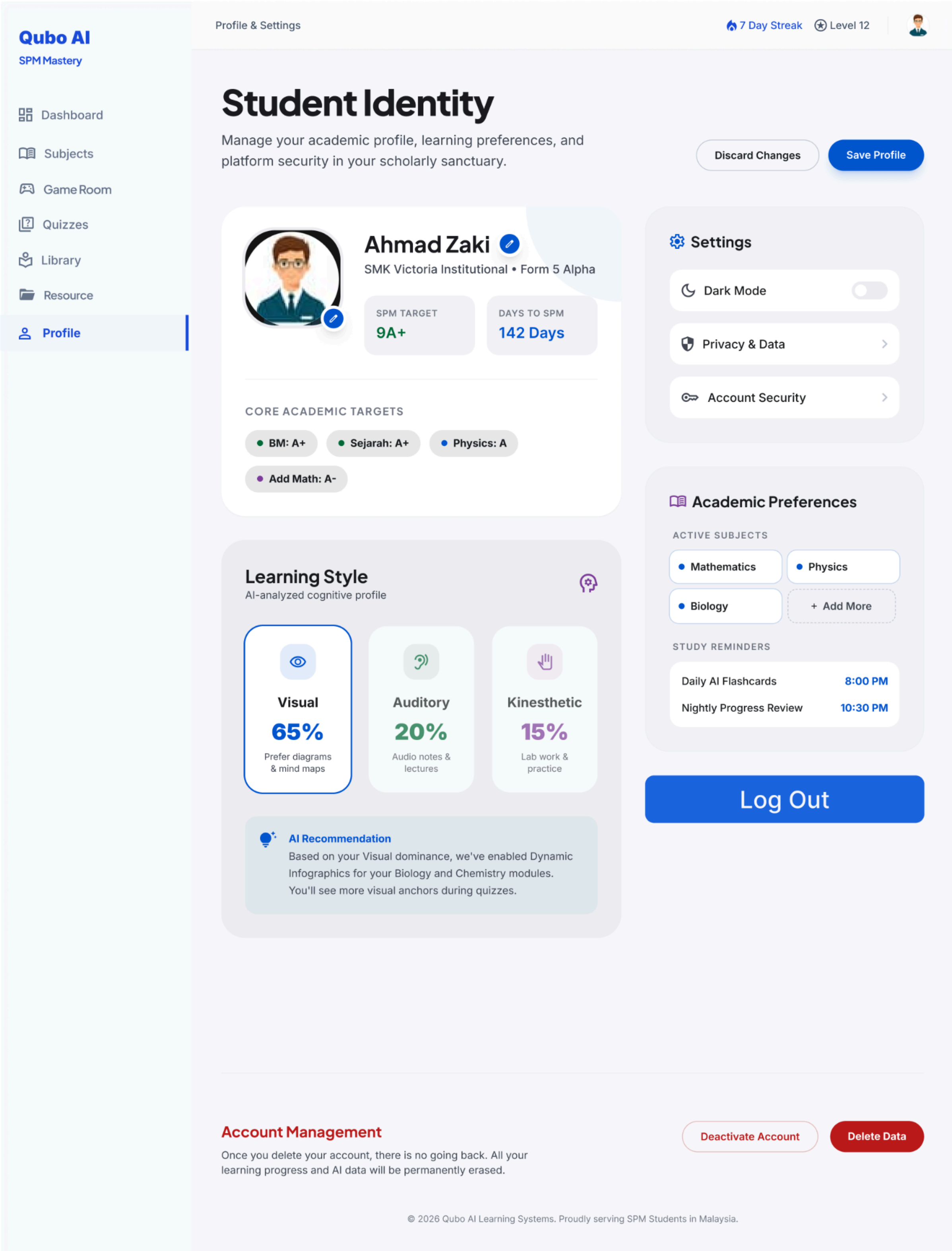


Figure 4.7: Profile and Setting Page

4.2 Use Case Diagram

4.2.1 User Authentication and Profile Management Module

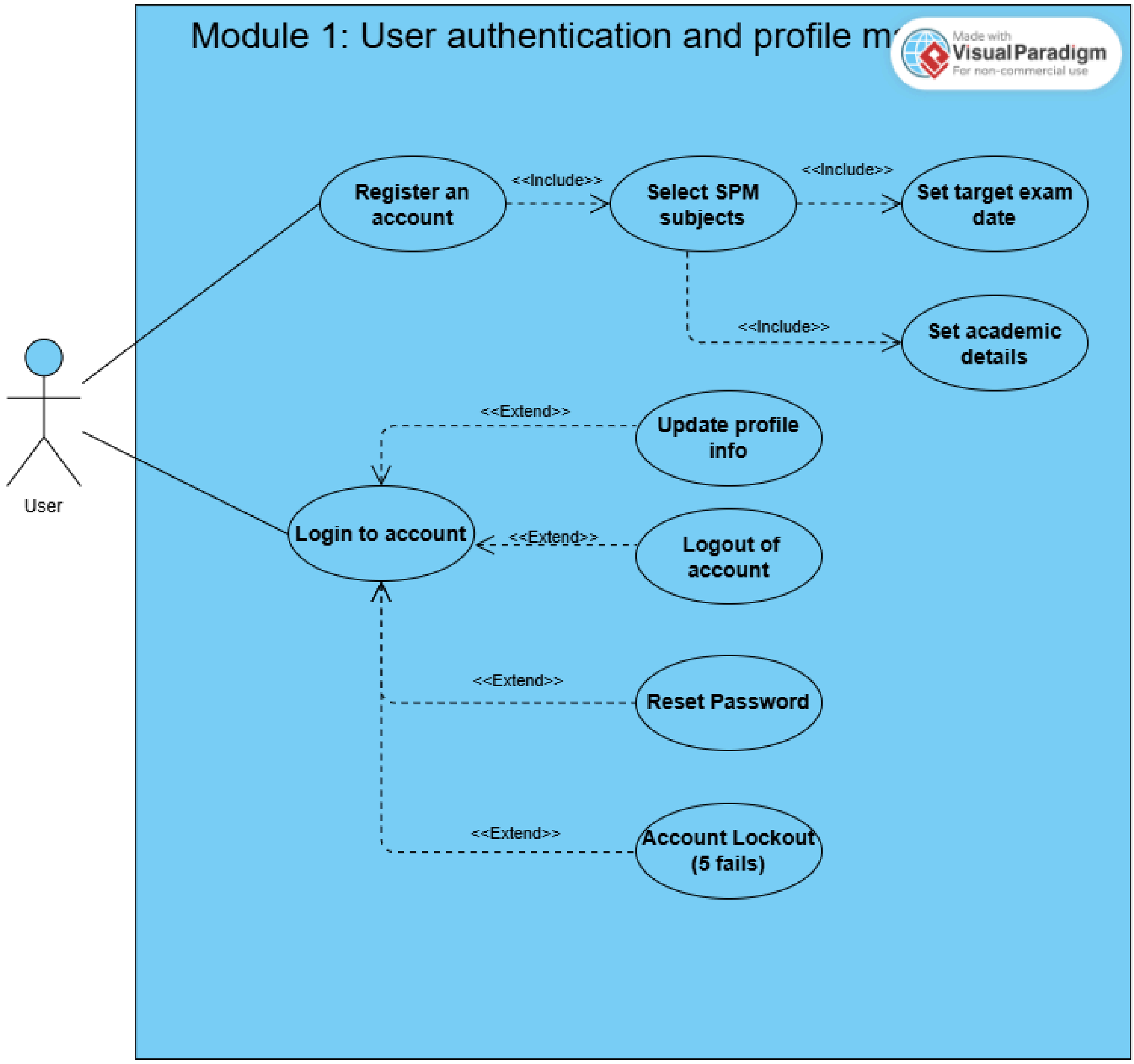


Figure 4.8: Use case of User Authentication and Profile Management Module

Table 4.1: Use Case Description of Module 1

Use Case Name	UC100 User Registration
Objectives	Allow a new user to create a Qubo account by providing their personal details and role (Student / Educator).
Actor / Activator	Unregistered User, System
Pre-Conditions	1. The registration page is accessible. 2.The user does not already have an account linked to the provided email.
Basic Flows	
Actor	**System**
1. The user navigates to the registration page.	2. The system displays a registration form with fields: Full Name, Email, Password, Confirm Password, and Role.
3. The user fills in all required fields and selects a role.	4. System validates input in real-time [A1: Invalid Input]
5.. The user clicks the 'Register' button.	6. System checks for duplicate email addresses. [A2: Email Already Exists] 7. The system hashes passwords and creates user records in the database. 8. The system sends a verification email to the registered address. 9. The system displays a confirmation message [M1: Registration Success] 10. Redirects the user to the login page.
Alternate Flow	
A1 : Invalid Input	
A1.1 System detects missing or incorrectly formatted field(s). A1.2 System highlights the invalid field(s) and displays inline error messages [M2: Validation Error]. A1.3 Return to BF:2.	
A2: Email Already Exists	
A2.1 System detects that the email is already registered. A2.2 System displays an error message [M3: Email Taken]. A2.3 Return to BF:2.	
Messages [M]	

M1: Msg Registration Success = {'Account created successfully!' } M2: Err Validation = { 'Password must be at least 8 characters and contain one uppercase letter, one number, and one symbol.'} M3: Msg Email Taken = {' An account with this email address already exists. Please log in or use a different email'}	
Post-Conditions	1. User is registered to the system

Table 4.2: Use Case Description of Module 1

<table>
<tr><td>Use Case Name</td><td colspan="2">UC200 User Login</td></tr>
<tr><td>Objectives</td><td colspan="2">Allow a registered and verified user to authenticate and access their personalised dashboard.</td></tr>
<tr><td>Actor / Activator</td><td colspan="2">Registered User, System</td></tr>
<tr><td>Pre-Conditions</td><td colspan="2">1. The user has a verified Qubo account..
2. The login page is accessible.</td></tr>
<tr><td colspan="3">Basic Flows</td></tr>
<tr><td colspan="2">Actor</td><td>System</td></tr>
<tr><td colspan="2">1. The user navigates to the login page.</td><td>2. The system displays login form with Email and Password fields.</td></tr>
<tr><td colspan="2">3. The user enters their registered email and password.</td><td>4. System validates input format [A1: Invalid Format]</td></tr>
<tr><td colspan="2">5.. The user clicks the 'Login' button.</td><td>6. System verifies credentials against the database [A2: Wrong Credentials] [A3:Account Not Verified]

7. The system generates a session token and records login timestamp.

8. System redirects users to their role-specific dashboard (Student / Educator / Admin).

9. The system displays a personalized welcome message [M1: Welcome Back].</td></tr>
<tr><td colspan="3">Alternate Flow</td></tr>
<tr><td colspan="3">A1 : Invalid Format</td></tr>
<tr><td colspan="3">A1.1 System detects malformed email or empty fields.
A1.2 System displays inline error [M2: Invalid Format].
A1.3 Return to BF:2.</td></tr>
<tr><td colspan="3">A2: Wrong Credentials</td></tr>
<tr><td colspan="3">A2.1 System detects email/password mismatch.
A2.2 System increments failed-login counter.
A2.3 System displays an error message [M3: Wrong Credentials].
A2.4 If counter >= 5, the system locks the account temporarily [M4: Account Locked] and sends an unlock email.
A2.5 Return to BF:2.</td></tr>
<tr><td colspan="3">Messages [M]</td></tr>
</table>

M1: Msg Welcome Back = {'Welcome back, [Name]! Ready to continue learning?' } M2: Err Invalid Format = { 'Please enter a valid email address.'} M3: Err Wrong Credentials = {‘ 'Incorrect email or password. Please try again’} M4: Msg Account Lock = {'Your account has been temporarily locked due to multiple failed login attempts. Please check your email.'}	
Post-Conditions	1. The user is authenticated and a valid session is active. 2. The user is directed to the appropriate role-based dashboard.

4.2.2 Performance Tracking and Analytics Dashboard Module

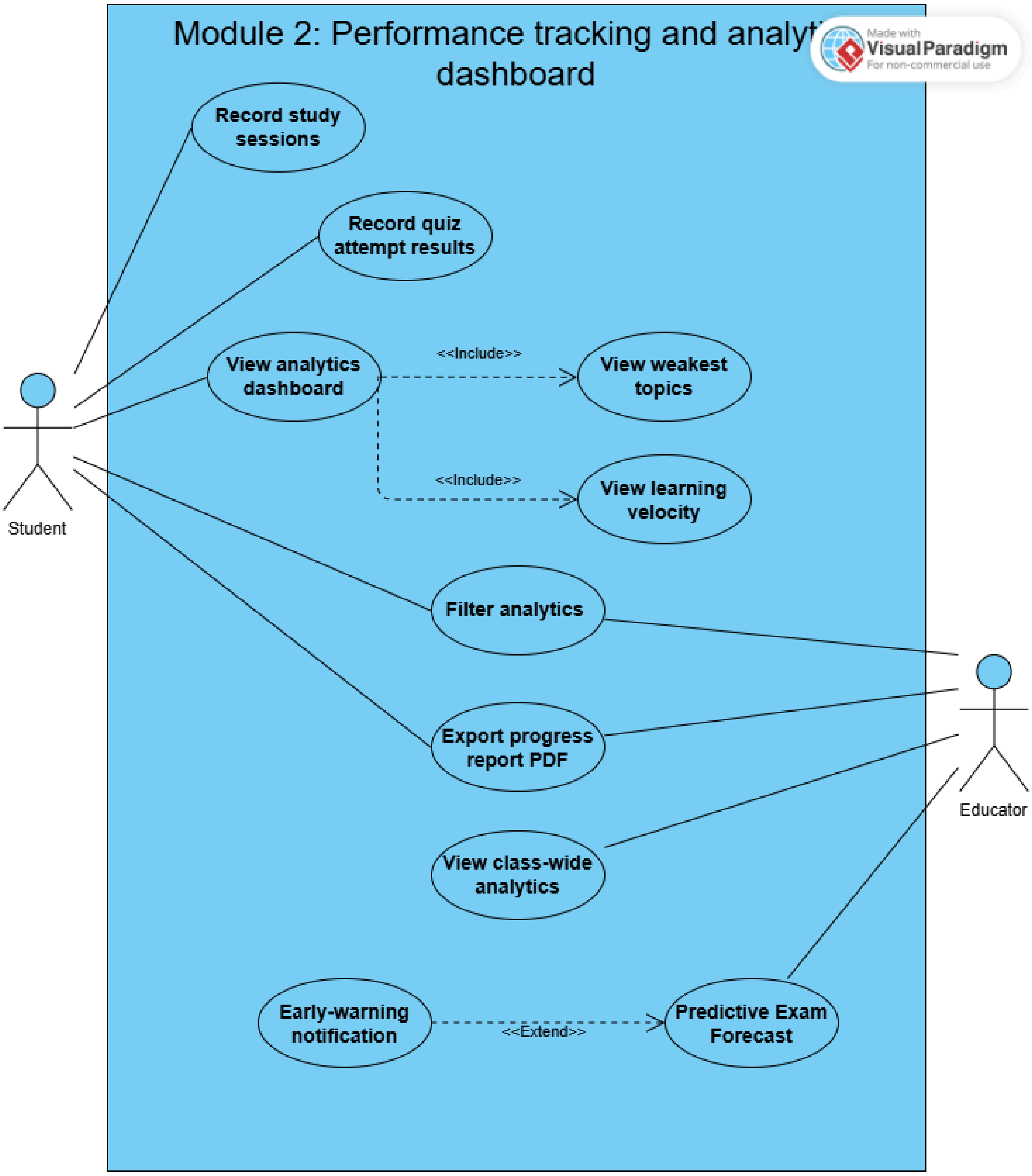


Figure 4.9: Use case of Performance Tracking and Analytics Dashboard Module

Table 4.3: Use Case Description of Module 2

Use Case Name	UC300 Log Study Session
Objectives	Allow a Student to record a study session including subject, duration (via Pomodoro timer), and self-rated difficulty to build their learning profile.
Actor / Activator	Student (Registered User), System
Pre-Conditions	1. Student is logged in. 2. Student has at least one subject configured in their profile.
Basic Flows	
Actor	**System**
1. The student navigates to the 'Study Room'.	2. The system displays the study session interface with subject selector and Pomodoro timer.
3. The student selects a subject and topic.	4. The system loads topic list for the selected subject from the curriculum database.
5.. Students click 'Start Session'.	6. The system starts the Pomodoro timer (default: 15 minutes) and begins recording session data.
7. Student studies; timer runs in foreground.[A1: Session Interrupted]	8. The system counts down and triggers short break reminders at Pomodoro intervals. [M1: Break Reminder]
9. Students click 'End Session'.	10. System stops the timer and records total duration.
11. Students rate the session difficulty (Easy / Medium / Hard).	12. The system captures difficulty rating and saves the complete session record. 13. System updates the student's learning profile, streak counter, and gamification points. 14. System displays session summary. [M2: Session Saved]
Alternate Flow	
A1 : Session Interrupted	
A1.1 Student closes the browser or loses internet connection mid-session. A1.2 System auto-saves partial session data upon reconnection. A1.3 System prompts students to resume or discard. [M3: Resume Session] A1.4 Return to BF:9.	
Messages [M]	

M1: Msg Break Reminder: = { 'Great work! Time for a 5-minute break. You have completed [N] Pomodoros today.' } M2: Msg Session Saved: = { 'Session recorded: [Subject] – [Duration] mins. Keep it up! You have earned [X] XP.' } M3: Msg Resume Session: = { 'It looks like your last session was interrupted. Would you like to resume or start fresh?' }	
Constraint[C]	
C1: Session Minimum A session must have a minimum duration of 5 minutes to be counted towards analytics and streak tracking.	
Post-Conditions	1. Study session record is persisted in the database. 2. Student's learning profile, streak, and XP are updated accordingly.

Table 4.4: Use Case Description of Module 2

Use Case Name	UC400 View Performance Analytics Dashboard
Objectives	Allow a Student to view comprehensive learning analytics including subject performance, study trends, predicted exam scores, and AI-generated study recommendations.
Actor / Activator	Student (Registered User), AI Processing Engine, System
Pre-Conditions	1. The student is logged in. 2. Students has at least one completed study session or quiz attempt recorded.
Basic Flows	
Actor	**System**
1. Student navigates to 'My Dashboard'.	2. System retrieves all study session records, quiz scores, and engagement data for the student. 3. AI Engine processes data and computes performance metrics (subject scores, learning velocity, weak topics). 4. System renders analytics dashboard with interactive charts: subject-wise performance, study time trends, and anonymised class average comparison.

	5. System displays Predicted Exam Score widget with confidence interval for each SPM subject. [C1: Prediction Algorithm] 6. AI Engine generates personalised study recommendations. [C2: Recommendation Logic] 7. System displays at-risk alerts for subjects with predicted performance below 60%. [M1: At-Risk Alert]
8. Student clicks on a subject for a drill-down view.	9. System loads subject-specific analytics: topic breakdown, time invested, and quiz history.
10. Student clicks 'Generate Study Plan'.	11. AI Engine creates a weekly study plan based on weaknesses and upcoming exam date. [A1: Insufficient Data] 12. System displays generated study plan. [M2: Study Plan Ready]
Alternate Flow	
A1: Insufficient Data A1.1 AI Engine determines data points are insufficient for reliable predictions (< 3 sessions). A1.2 System displays placeholder with guidance. [M3: More Data Needed] A1.3 Skip to BF:8.	
Messages [M]	
M1: Msg At-Risk Alert: = { 'Warning: Based on current performance, you are at risk of underperforming in [Subject]. Consider increasing study time and reviewing flagged topics.' } M2: Msg Study Plan Ready: = { 'Your personalised study plan for the next 7 days has been generated. View it under Study Plans.' } M3: Msg More Data Needed: = { 'Complete at least 3 study sessions to unlock full analytics and predictions.' }	
Constraint [C]	

C1: Prediction Algorithm Predicted Score = (Quiz Score Weight x 0.5) + (Study Time Weight x 0.3) + (Recency Weight x 0.2) Confidence increases with more data points; displayed as a percentage alongside the predicted score. Predictions are recalculated after each new quiz attempt or study session. **C2: Recommendation Logic** Topics with quiz accuracy < 60% are flagged as high-priority review items. Recommendations include: videos (YouTube), articles, or practice problems matched to the student's learning style (Visual / Auditory / Kinesthetic). Maximum of 5 recommendations displayed per session to avoid overwhelming the student.
Post-Conditions
1. Dashboard is rendered with up-to-date analytics. 2. Study plan (if generated) is saved to the student's profile.

Table 4.5: Use Case Description of Module 2

Use Case Name	UC500 Monitor Student Progress (Educator)
Objectives	Allow an Educator to monitor class-wide and individual student performance, identify at-risk learners, and access aggregated analytics to inform teaching strategies.
Actor / Activator	Educator (Registered User), System
Pre-Conditions	1. Educator is logged in with the 'Educator' role. 2. At least one student has been enrolled in the educator's class.
Basic Flows	
Actor	**System**
1. Educator navigates to 'Class Dashboard'.	2. System retrieves aggregated performance data for all students in the educator's class. 3. System renders class-wide analytics: average scores by subject, participation rates, and study time distribution.

	4. System highlights at-risk students (predicted performance < 60%) with a red indicator. [M1: At-Risk Students]
5. Educator clicks on an individual student.	6. System loads the student's detailed performance profile: quiz scores, study sessions, topic weaknesses, and trend lines.
7. Educator optionally applies subject or date filters.	8. System refreshes analytics based on applied filters.
9. Educator clicks 'Export Report'.	10. System generates a PDF progress report for the selected student or the entire class. [A1: No Data Available] 11. System confirms report generation [M2: Report Ready] and provides a download link.
Alternate Flow	
A1: No Data Available A1.1 System detects no activity records for the selected student or time range. A1.2 System displays informational message. [M3: No Data] A1.3 Return to BF:5.	
Messages [M]	
M1: Msg At-Risk Students: = { 'The following students may need additional support: [Student Name(s)]. Click to view their profiles.' } M2: Msg Report Ready: = { 'Progress report has been generated. Click here to download.' } M3: Msg No Data: = { 'No activity data is available for the selected criteria. Encourage the student to log study sessions.' }	
Post-Conditions	
1. Educator has reviewed class and individual performance data. 2. Generated report (if requested) is available for download.	

4.2.3 AI-powered Quiz Generation and Practice Management Module

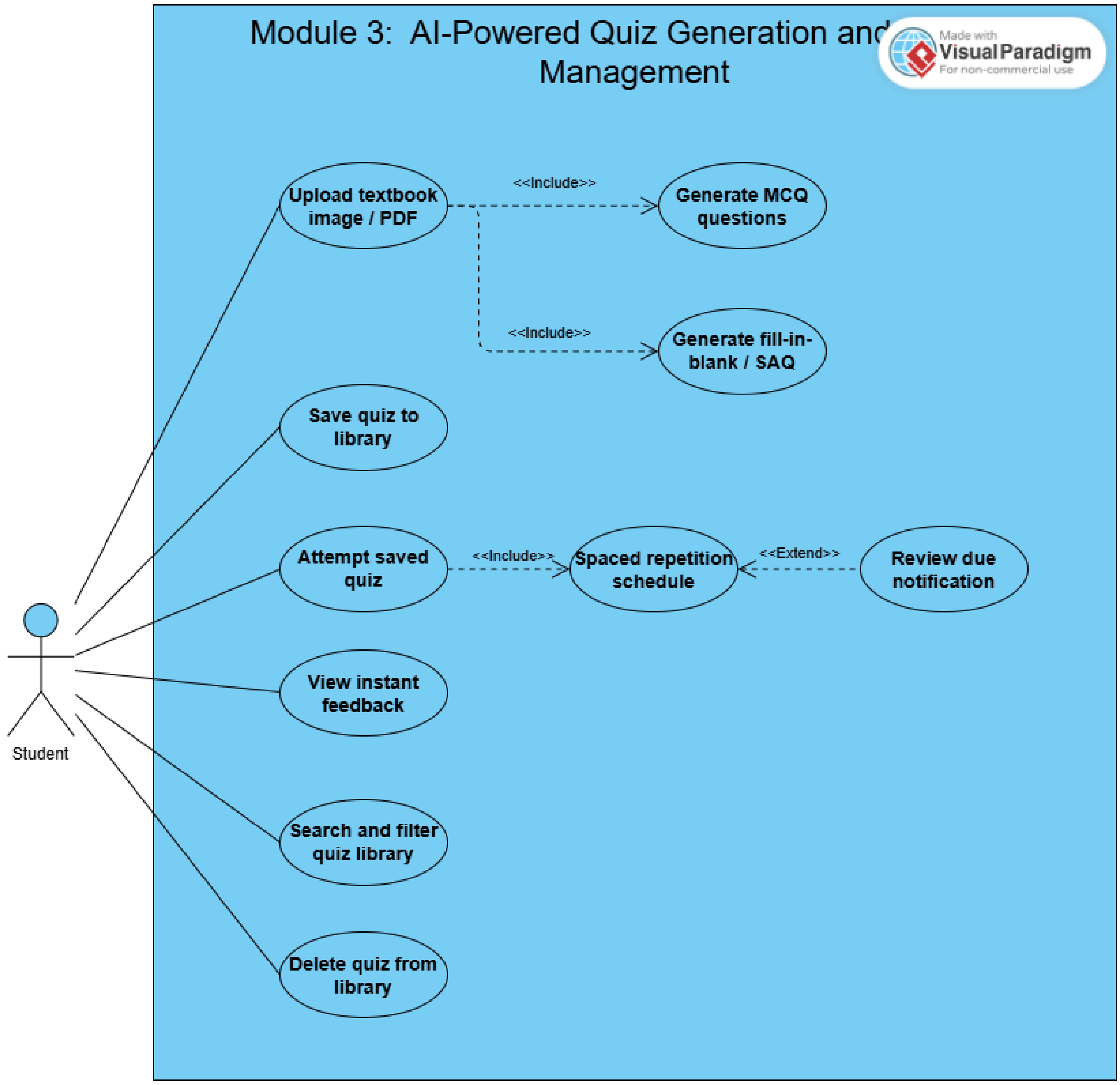


Figure 4.10: Use case of AI-powered Quiz Generation and Practice Management Module

Table 4.6: Use Case Description of Module 3

<table>
<tr><td>Use Case Name</td><td>UC600 Generate AI Quiz from Textbook Content</td></tr>
<tr><td>Objectives</td><td>Allow a Student to upload a textbook image or PDF page and have the AI engine auto-generate a quiz, which is then saved to the student's quiz library.</td></tr>
<tr><td>Actor / Activator</td><td>Student (Registered User), AI Processing Engine, System</td></tr>
<tr><td>Pre-Conditions</td><td>1. The student is logged in.
2. The AI NLP quiz generation service is operational.</td></tr>
<tr><td colspan="2">Basic Flows</td></tr>
<tr><td>Actor</td><td>System</td></tr>
<tr><td>1. Student navigates to 'Quiz Generator'.</td><td>2. System displays the upload interface.</td></tr>
<tr><td>3. Students upload an image (JPG/PNG) or PDF page from their textbook.</td><td>4. System validates file format and size. [A1: Unsupported Format] [A2: File Too Large]

5. System sends the uploaded content to the AI Processing Engine for text extraction and analysis.

6. AI Engine extracts key concepts, facts, and relationships from the content using NLP.</td></tr>
<tr><td>7. Student selects quiz type (MCQ / Short Answer / True-False) and number of questions (5–20).</td><td>8. System forwards configuration to AI Engine.

9. AI Engine generates quiz questions with answer keys.

10. System displays generated quiz for student preview. [M1: Quiz Ready] [A3: Poor Content Quality]</td></tr>
<tr><td>11. Student reviews, optionally edits questions, then clicks 'Save to Library'.</td><td>12. System saves the quiz to the student's personal quiz library.

13. System confirms save [M2: Quiz Saved] and offers option to attempt the quiz immediately.</td></tr>
<tr><td colspan="2">Alternate Flow</td></tr>
<tr><td colspan="2">A1: Unsupported Format</td></tr>
<tr><td colspan="2">A1.1 System detects unsupported file type.
A1.2 System displays error. [M3: Unsupported Format]
A1.3 Return to BF:3.</td></tr>
<tr><td colspan="2">A2: File Too Large</td></tr>
<tr><td colspan="2">A2.1 System detects files that exceed the 10 MB limit.
A2.2 System displays error. [M4: File Too Large]
A2.3 Return to BF:3.</td></tr>
<tr><td colspan="2">A3: Poor Content Quality</td></tr>
<tr><td colspan="2">A3.1 AI Engine determines insufficient extractable content (e.g., blurry image).</td></tr>
</table>

A3.2 System notifies student [M5: Low Quality Warning] and recommends re-upload. A3.3 Return to BF:3.	
Messages [M]	
M1: Msg Quiz Ready: = { 'Your quiz has been generated! Review the questions below before saving.' } M2: Msg Quiz Saved: = { '[Quiz Title] has been saved to your library. Start now?' } M3: Err Unsupported Format: = { 'Please upload a JPG, PNG, or PDF file.' } M4: Err File Too Large: = { 'File size must not exceed 10 MB. Please compress or crop the image and try again.' } M5: Err Low Quality Warning: = { 'The image quality is too low to extract meaningful content. Please upload a clearer image.' }	
Constraint[C]	
C1: Question Generation Minimum: 5 questions; Maximum: 20 questions per generation request. Supported quiz types: Multiple Choice (MCQ), Short Answer, True/False. AI uses semantic similarity >= 70% to avoid duplicate questions within the same quiz.	
Post-Conditions	1. Generated quiz is stored in the student's personal quiz library. 2. Quiz metadata (subject, topic, date created, question count) is indexed for analytics.

Table 4.7: Use Case Description of Module 3

Use Case Name	UC700 Attempt a Practice Quiz
Objectives	Allow a Student to attempt a saved or AI-recommended quiz and receive immediate feedback with performance scores.
Actor / Activator	Student (Registered User), System
Pre-Conditions	1. The student is logged in. 2. At least one quiz exists in the student's library or in the recommended quiz pool.
Basic Flows	
Actor	**System**

1. Student navigates to 'Quiz Library' or 'Recommended Quizzes'.	2. System displays available quizzes with metadata (subject, difficulty, estimated time).
3. Student selects a quiz and clicks 'Start Quiz'.	4. System loads quiz questions and starts a session timer.
5. Student answers each question sequentially.	6. System records each response and moves to the next question.
7. Student submits the quiz.	8. System evaluates responses against answer keys and calculates the score. [A1: Time Expired] 9. System updates spaced repetition schedule for topics based on performance. [C1: Spaced Repetition Update] 10. System awards XP and updates gamification metrics. 11. System displays detailed results page [M1: Quiz Results] with score, correct/incorrect breakdown, and explanations.
12. Student reviews results and can click 'Retry' or 'Back to Library'.	13. System logs quiz attempt to performance analytics.
Alternate Flow	
A1: Time Expired	
A1.1 Quiz timer reaches zero. A1.2 System auto-submits any answered questions. A1.3 System displays [M2: Time Expired] before showing results. A1.4 Skip to BF:9.	
A2: Exit Mid-Quiz	
A2.1 Student clicks 'Exit' during an ongoing quiz. A2.2 System prompts confirmation. [M3: Confirm Exit] A2.3 If confirmed, system discards the current attempt (no score recorded). A2.4 Return to quiz library.	
Messages [M]	

M1: Msg Quiz Results: = { 'You scored [X]% on [Quiz Title]. Review your answers below and see where to improve.' } M2: Msg Time Expired: = { 'Time is up! Your answers have been submitted automatically.' } M3: Msg Confirm Exit: = { 'Are you sure you want to exit? Your current progress will not be saved.' }
Constraint [C]
C1: Spaced Repetition Update Topics scored < 60%: scheduled for review within 1–3 days. Topics scored 60%–79%: scheduled for review within 5–7 days. Topics scored >= 80%: scheduled for review within 14 days.
Post-Conditions
1. Quiz attempt is persisted with score, timestamp, and per-question breakdown. 2. Spaced repetition schedule is updated for affected topics. 3. Student XP and gamification badges are updated.

4.3 Sequence Diagram

4.3.1 User Authentication and Profile Management Module

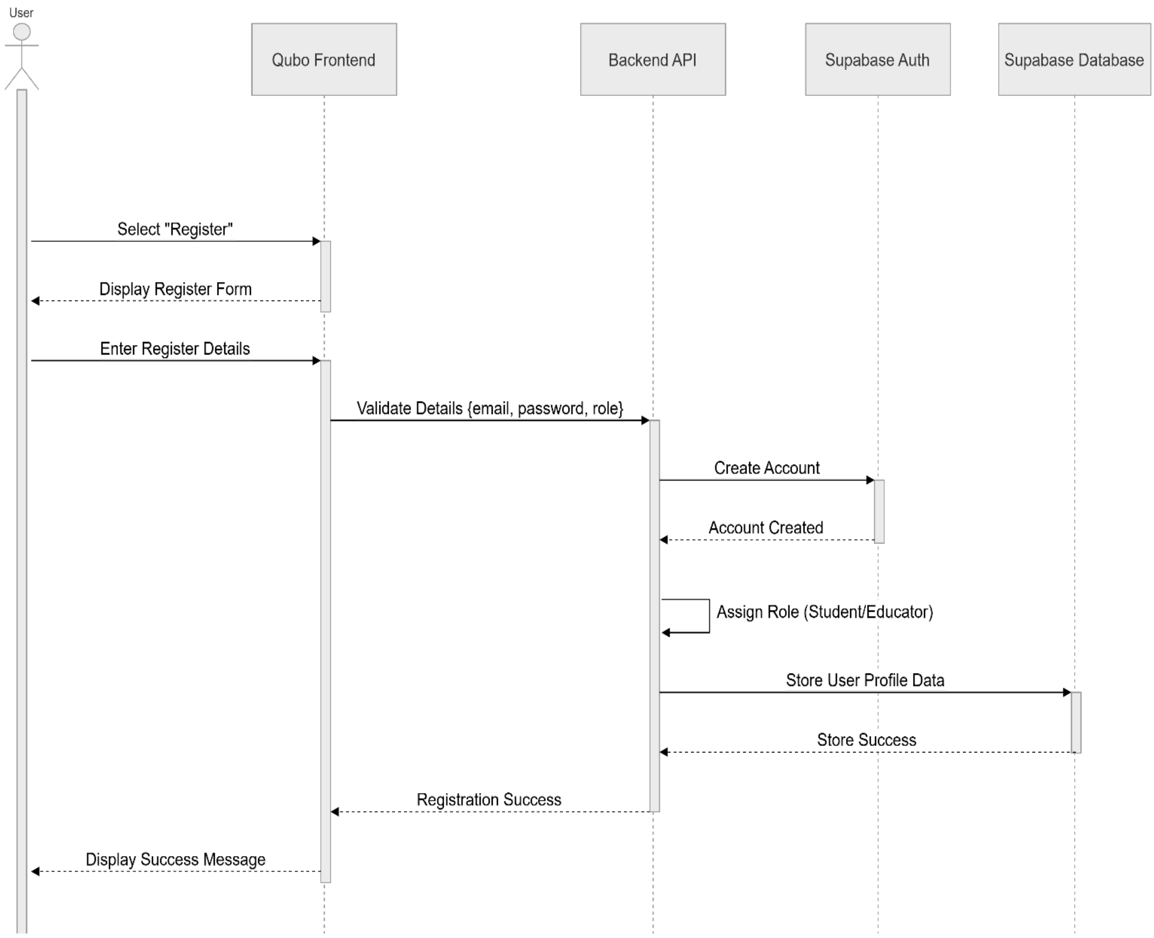


Figure 4.11: Sequence Diagram of User Authentication and Profile Management Module

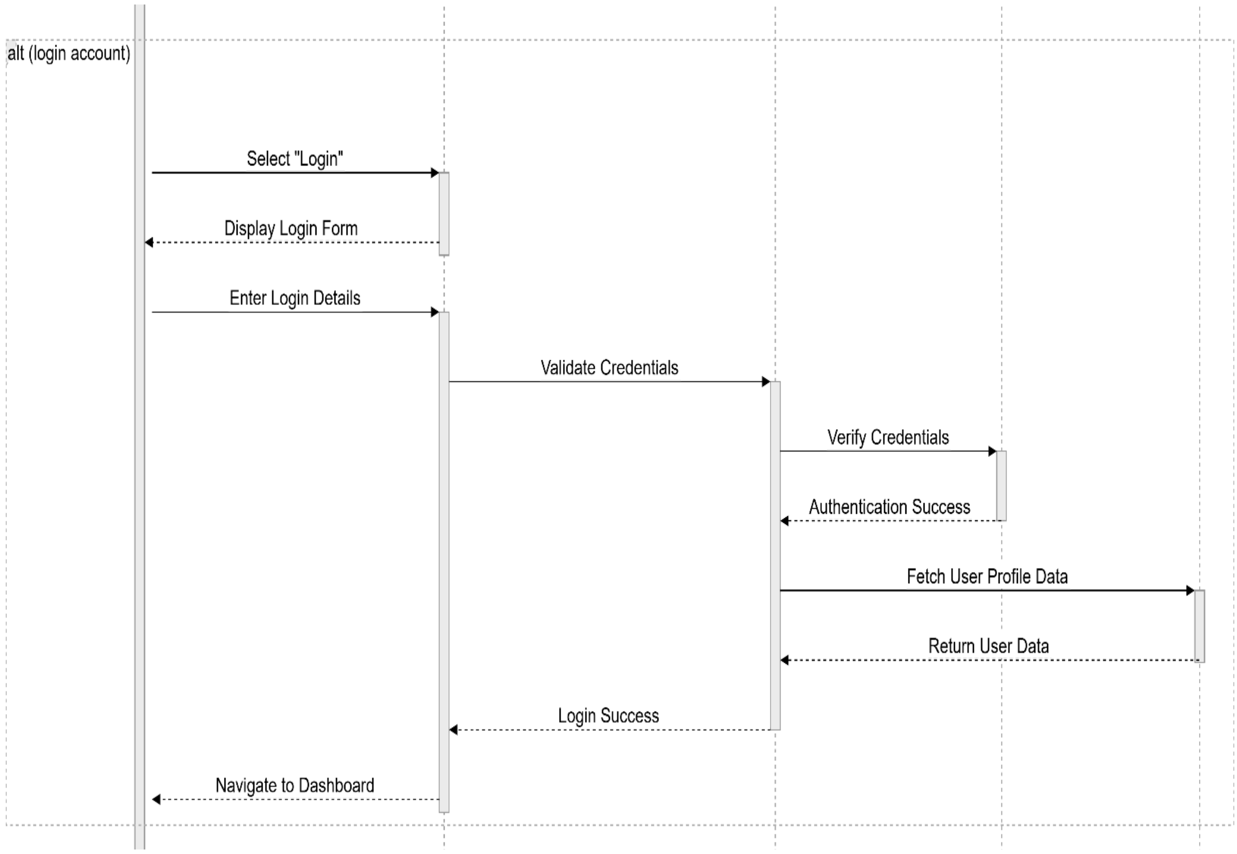


Figure 4.12: Sequence Diagram of User Authentication and Profile Management Module

4.3.2 Performance Tracking and Analytics Dashboard Module

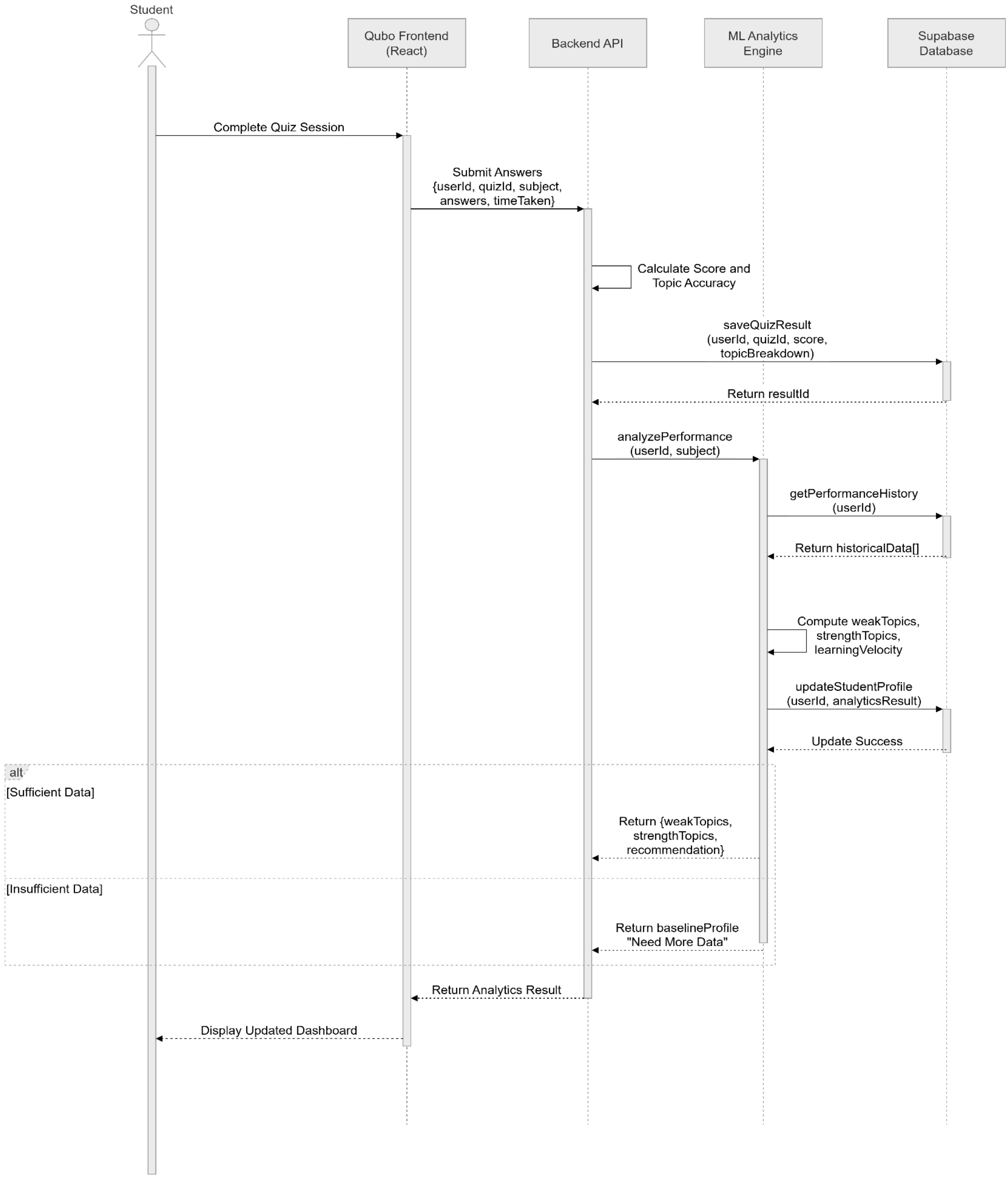


Figure 4.13: Sequence Diagram of Performance Tracking and Analytics Dashboard Module

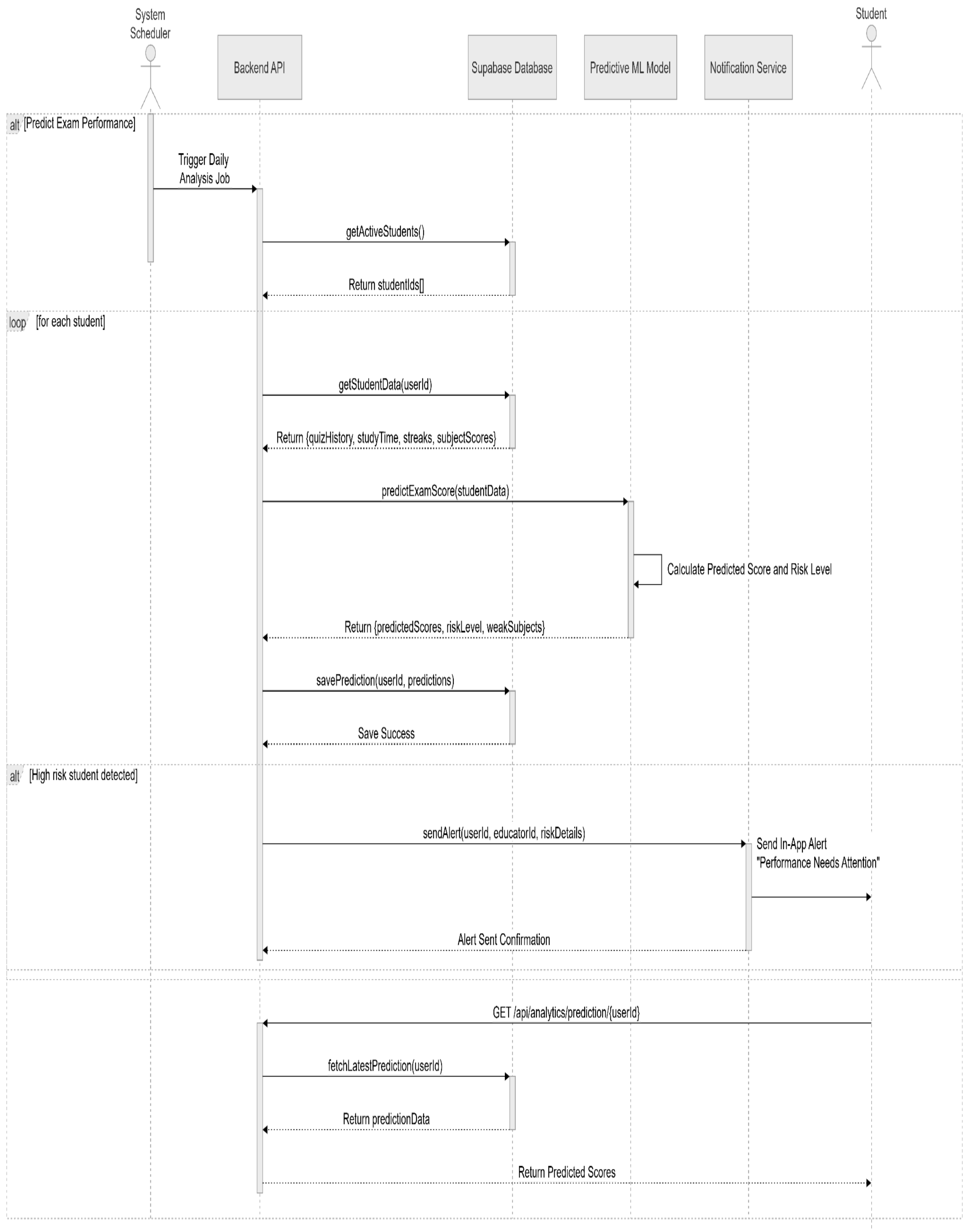


Figure 4.14: Sequence Diagram of Performance Tracking and Analytics Dashboard Module

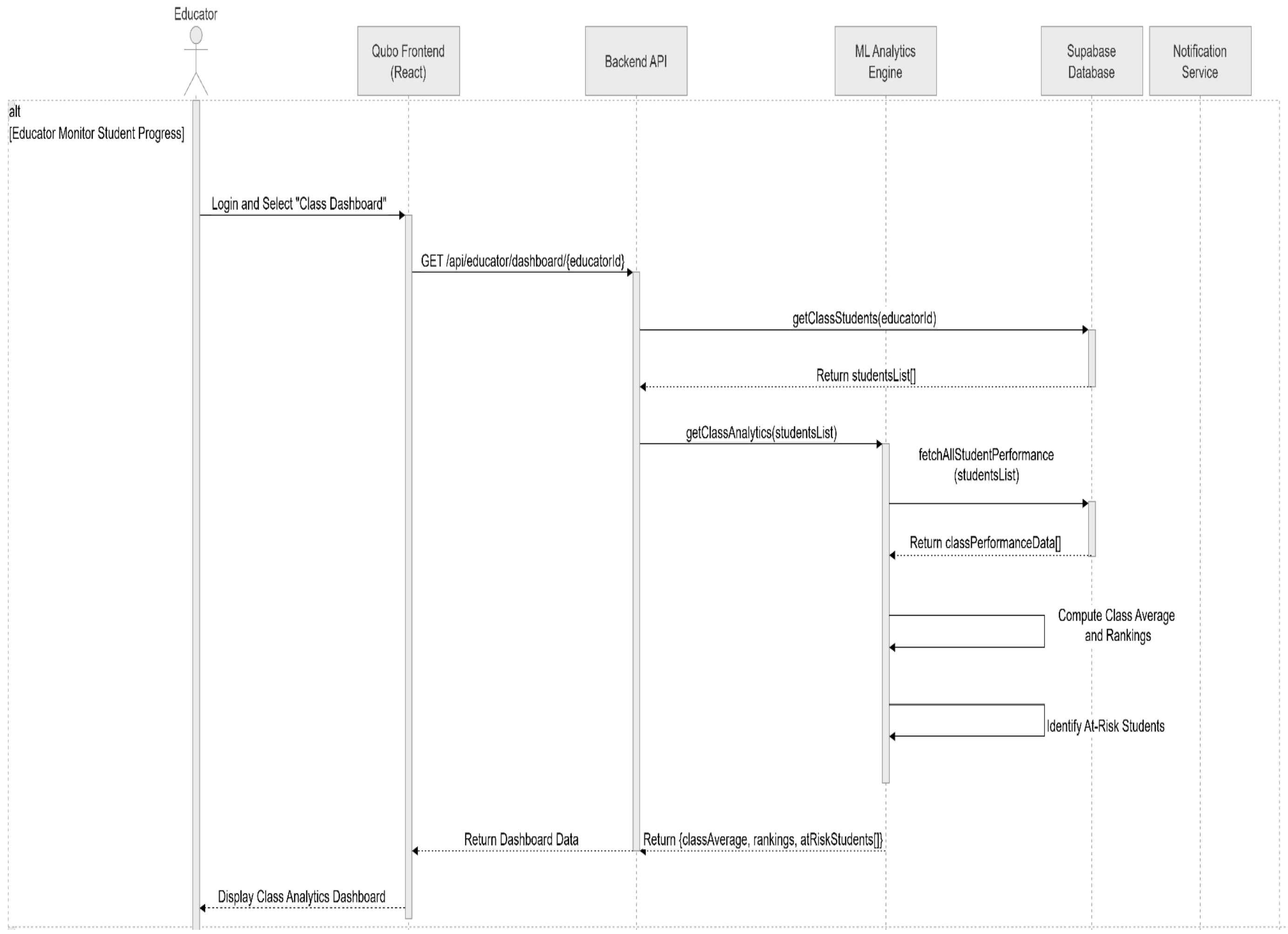


Figure 4.15: Sequence Diagram of Performance Tracking and Analytics Dashboard Module

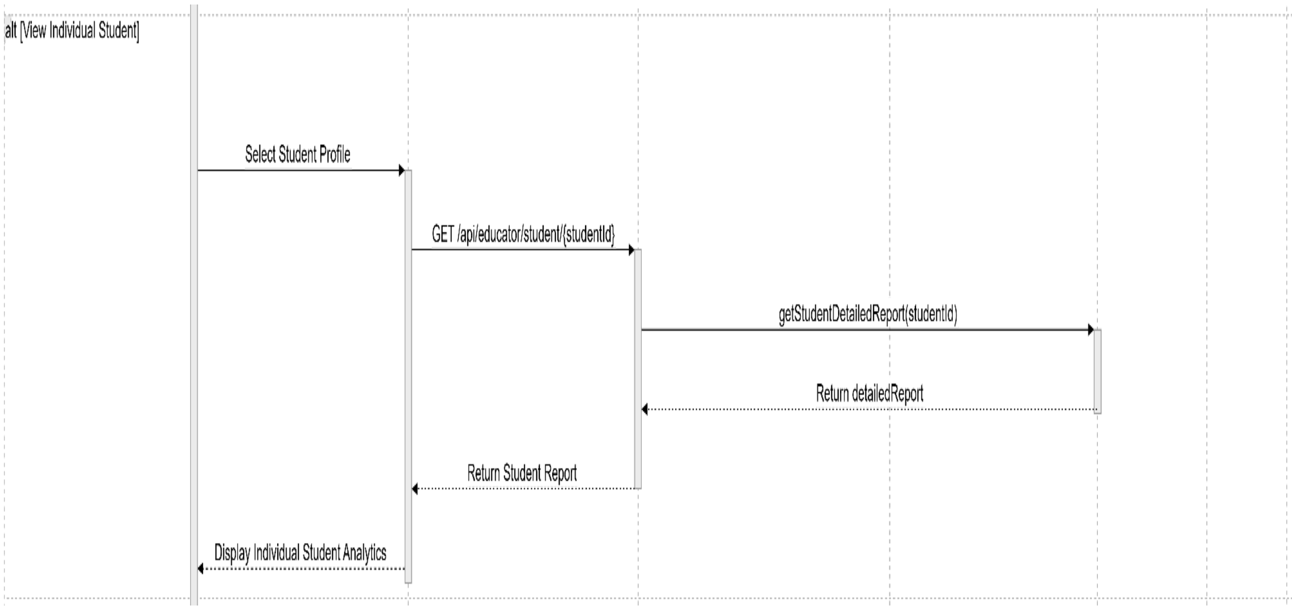


Figure 4.16: Sequence Diagram of Performance Tracking and Analytics Dashboard Module

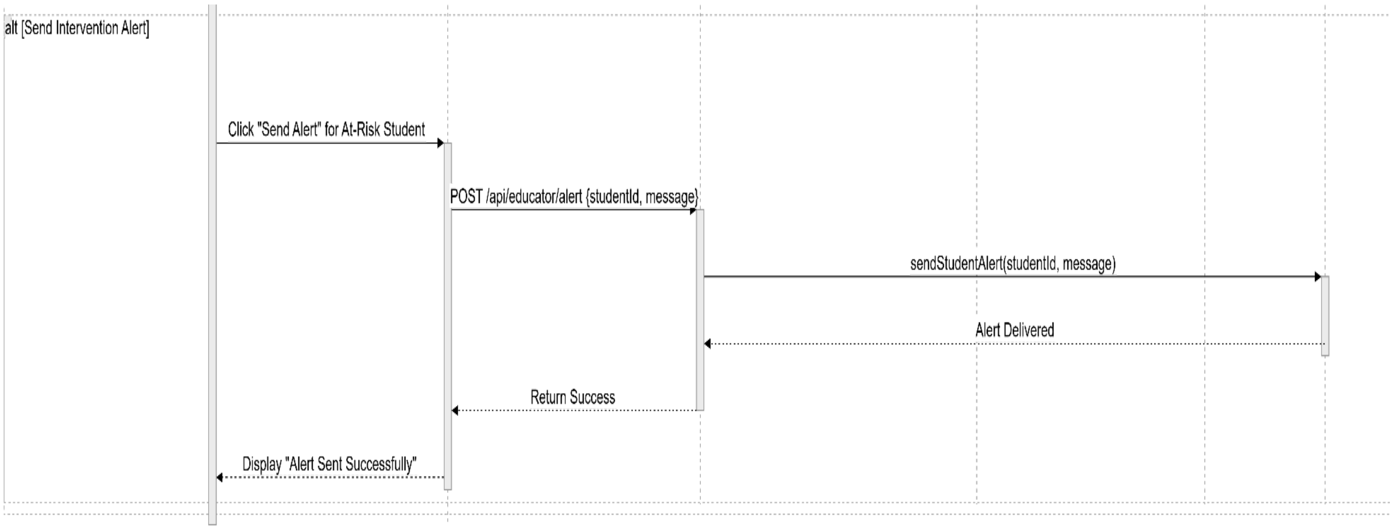


Figure 4.17: Sequence Diagram of Performance Tracking and Analytics Dashboard Module

4.3.3 AI-powered Quiz Generation and Practice Management Module

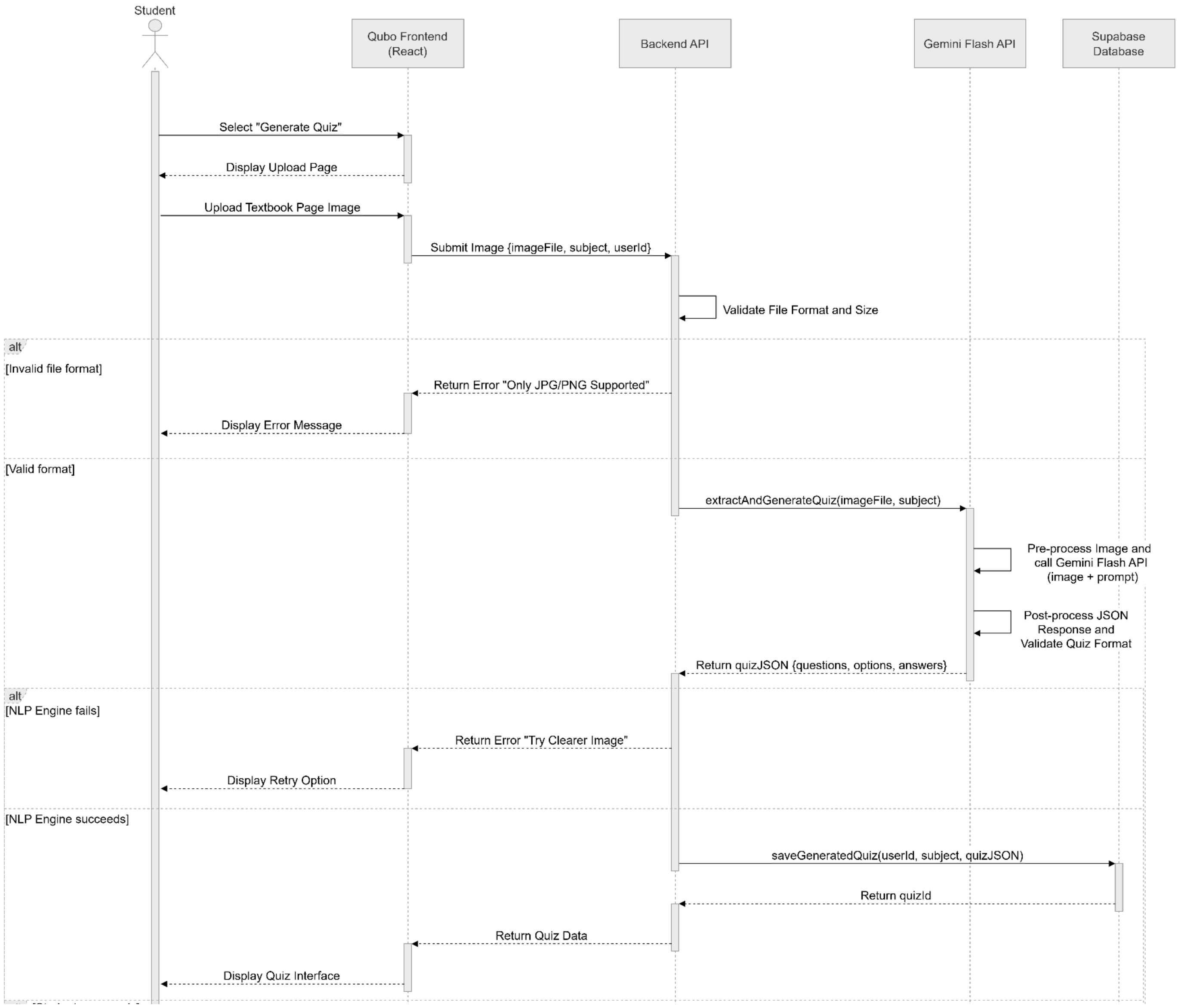


Figure 4.18: Sequence diagram of AI-powered Quiz Generation and Practice Management Module

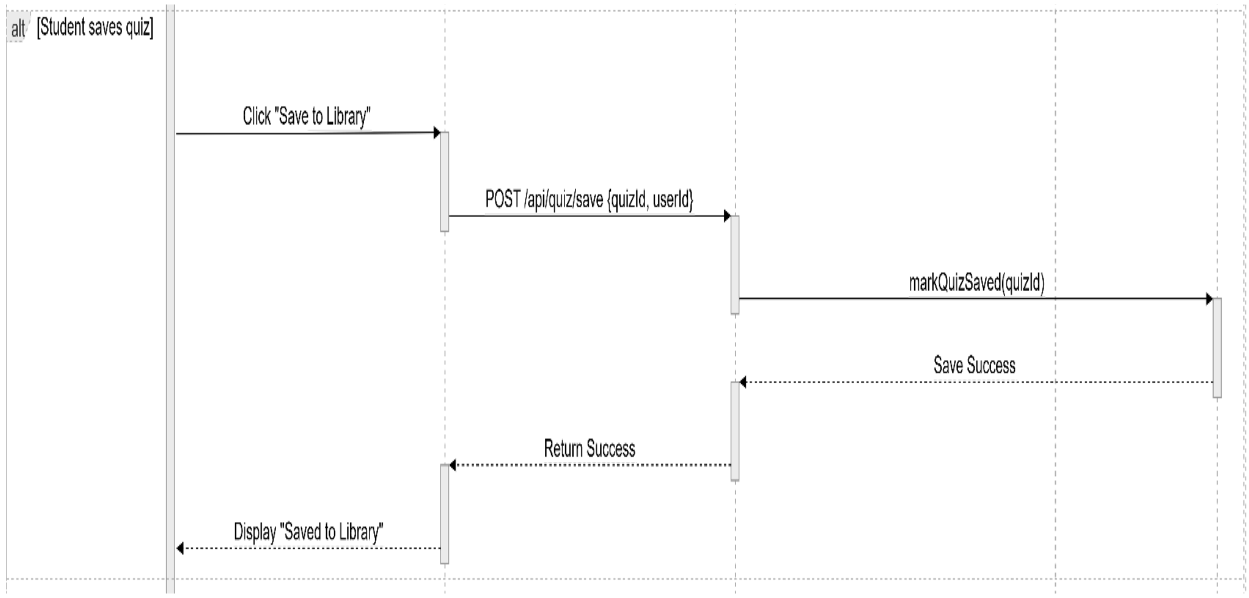
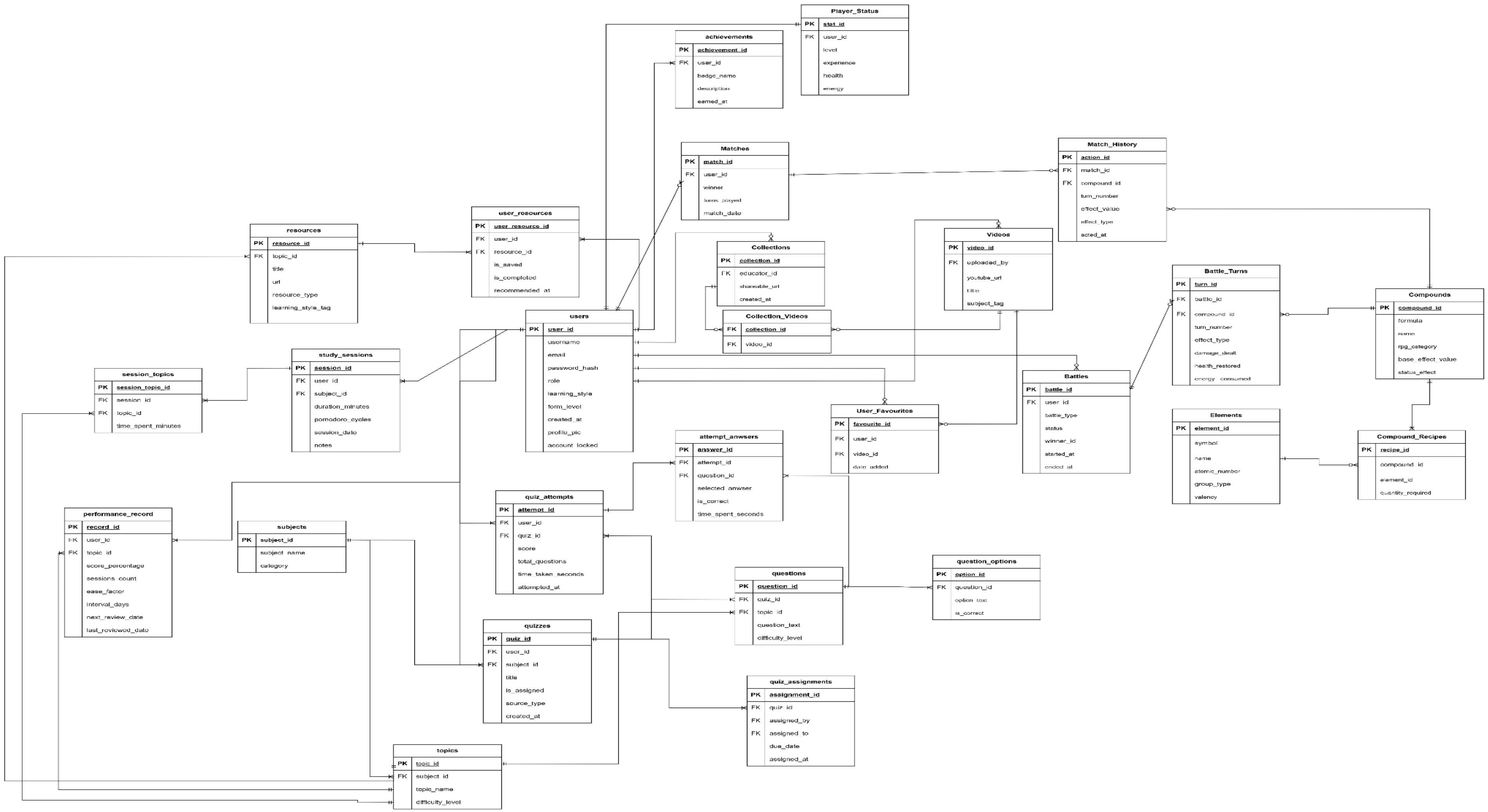
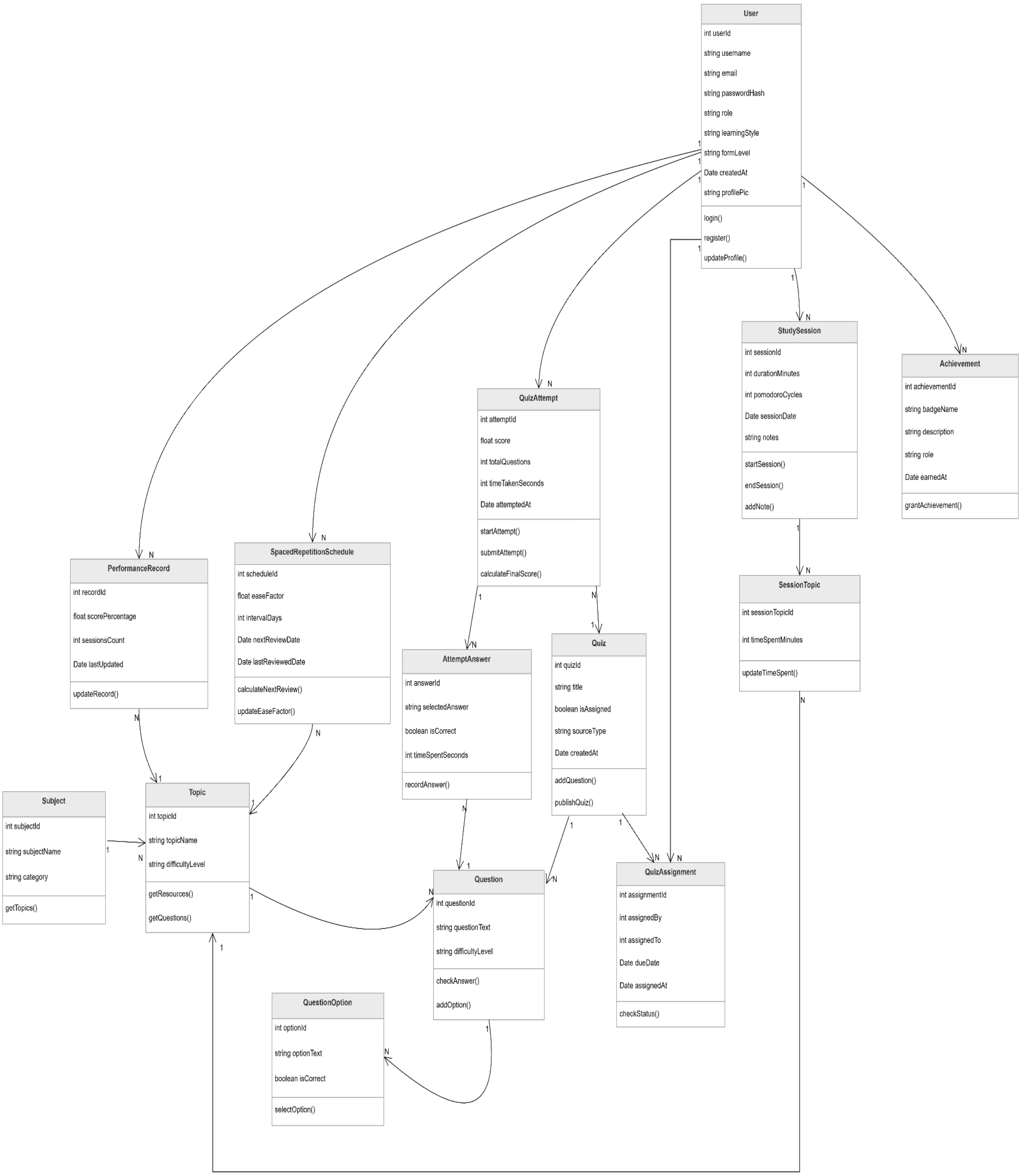


Figure 4.19: Sequence diagram of AI-powered Quiz Generation and Practice Management Modul

4.4 Entity Relationship Diagram



4.5 Class Diagram



4.6 Deployment Diagram

4.7 Chapter Summary and Evaluation

This chapter presented the complete system design for Qubo, the AI-Powered Personalized Learning Platform. The chapter was structured across six sections, covering the user interface design, use case diagrams, sequence diagrams, entity relationship diagram, class diagram, and deployment diagram.

Section 4.1 introduced the User Interface designs, showcasing wireframes for the Student Dashboard, Educator Dashboard, Performance Tracking and Analytics Page, AI Quiz Generation Page, Quiz Experience, Library Page, and Profile and Settings Page. These mockups illustrate the overall look and flow of the application, ensuring a student-centred and intuitive user experience.

Sections 4.2 through 4.3 documented the system's behavioural design. Use case diagrams were produced for the three core modules that are User Authentication and Profile Management, Performance Tracking and Analytics Dashboard, and the AI-Powered Quiz Generation and Practice Management module. To define the interactions between users and the system. Sequence diagrams then elaborated on the step-by-step message flow within each module, particularly capturing the multi-step interactions involved in analytics processing and AI quiz generation.

Sections 4.4 to 4.6 addressed the structural and deployment aspects of the system. The Entity Relationship Diagram defined the data model and relationships between key entities such as users, quizzes, sessions, and performance records. The Class Diagram outlined the internal object structure of the application. The Deployment Diagram illustrates how the system's components including the React frontend, Python/AI backend, and Supabase database, are distributed across the deployment environment.

Collectively, the design artefacts in this chapter provide a comprehensive blueprint for the implementation phase. The designs confirm that all functional and non-functional requirements defined in Chapter 3 are architecturally supported, and they serve as the foundation for the development work to be carried out in Project II.

Chapter 5

Implementation and Testing

5 Implementation and Testing

A short introduction that describes what will be included in this chapter.

IMPORTANT NOTE TO STUDENTS: Include details about:

- **Implementation**

  A detailed description of how you actually carried out the implementation (e.g., coding, etc.) of your system/prototype.

  - Include code snippets and descriptions to show how the requirements of the application/prototype have been met –
    - For Smart Campus projects, code snippets of the MQTT protocol for both client side and server side has to be included with explanation.
  - Include descriptions of important settings - Setting for server, network protocol, IP, database, security
  - Note: this should **not** be a chronological account of the work you carried out.

- **Testing**

  All test cases that have been carried out must be provided - To tabulate the test cases in tables

Note: Students may also opt to split Implementation and Testing into 2 separate chapters.

5.1 Sub-section 1 Heading

Sub-sections should be used to divide the chapter into logical parts.

5.1.1 Sub-subsection Heading

Sub-section numbering should be limited to a maximum of 3 levels (e.g. 5.3.1) in order to avoid confusion.

5.2 Sub-section 2 Heading

5.3 Sub-section 1 Heading

Sub-sections should be used to divide the chapter into logical parts.

5.3.1 Sub-subsection Heading

Sub-section numbering should be limited to a maximum of 3 levels (e.g. 5.3.1) in order to avoid confusion.

5.4 Sub-section 1 Heading

Sub-sections should be used to divide the chapter into logical parts.

5.4.1 Sub-subsection Heading

Sub-section numbering should be limited to a maximum of 3 levels (e.g. 5.3.1) in order to avoid confusion.

5.5 Chapter Summary and Evaluation

At the end of each chapter, evaluate the contents stated or discussed in the relevant sub-sections.

Chapter 6 *(if applicable)*

System Deployment

6 System Deployment

This chapter would be applicable for students who have embarked on a real-life industrial project. Students in this case would need to describe how the deployment has been carried out. Some of implementation tasks which need to be described include: training, file conversion or creation, and changeovers.

System Backup and Risk Management

Describe the procedures to backup the existing system for changeover purpose. Discuss the potential risk(s) for the changeovers and the solution.

On-site Setup

Describe the preparations to be done prior to the setup of the new system on client's site. Discuss the procedures to setup the new system, schedule, etc.

Training Procedure

Describe the procedures on the training procedures, contents, schedule, etc.

Follow-up

Describe the plan or procedures to follow up with the client (company) to verify the system reliability.

Chapter Summary and Evaluation

At the end of each chapter, evaluate the contents stated or discussed in the relevant sub-sections. For example,

6. Problems faced. Describe the various problems faced by students in the course of doing the project.
7. Solutions. What have been done to solve the problems?
8. What tools and techniques have been used and reasons for using them.

Chapter 7

Discussions and Conclusion

7 Discussions and Conclusion

Each student is required to make an *evaluation of the project* he/she has embarked on. The project evaluation may include the following sections.

IMPORTANT NOTE TO STUDENTS: In this chapter, for problems related to code, hardware, internet connection, etc (where applicable):

- List the technical problems faced and state how they were resolved
- List the unsolved technical problems for future enhancement
- List the achieved objectives/modules
- List the incomplete parts for future enhancement - this is regardless the parts listed in the pre-determined scope. Suggestions for future improvement

Summary

Summarize the project including the problem and proposed solutions, justification of the choice of tools, techniques and methodologies used in this project.

Achievements

Students are required to evaluate the project's achievement against project objectives, completion of the project, students' view of the strengths and weaknesses of the work done.

Contributions

Discuss the creativity, innovativeness, contribution of the proposed system. Explain why the proposed system is necessary. Describe the marketability of the system.

Limitations and Future Improvements

Identify the limitations of the research or project. Provide suggestions for improvement or further development of the system or research in the future.

Issues and Solutions

Students are required to describe the various problems faced by students during the project development and explain what has been done to solve the problems. The valuable experiences gained or lessons learnt through the project as a whole. The issues may include technical issues, project management issues, team dynamics problems, and other difficulties encountered and lessons learnt. How the issues are solved or can be solved to ensure the project can be completed on time or to be improved in the future.

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Appendices

In order to enhance better understanding of the project, students should as far as possible include all directly relevant materials, figures or diagrams in the **main body** rather than in the Appendix. The appendix is reserved only for items which may not directly be relevant or essential to enhance a reader's understanding of the project, or which may interrupt the smooth reading of the project document (for example being too voluminous).

Appendices should only include supportive materials **directly referred** to in the writing and should be kept to a **minimum**, e.g. selected pages of an annual report, not the entire document. Examples of items included in Appendices are:

- Company's report and documentation, such as sample invoice, purchase order form, etc.
- Project meeting documentation e.g. minutes of meetings, tracking documents, memos etc.
- Questionnaires and results, interview questions and results, pilot test and results, observation sheet and results, experiment test plan and results, etc.
- Analysis/design diagrams (only those not incorporated in the main body of the report).

If there is more than one appendix, they should be identified as A, B, etc (e.g. Appendix A). Formulae and equations in appendices should be given separate numbering: Eq. (A.1), Eq. (A.2), etc.; in a subsequent appendix, Eq. (B.1) and so on. Similarly for tables and figures: Table A.1; Fig. A.1, etc.

IMPORTANT NOTE TO STUDENTS

APPENDIX *n* User Guide

- List the username and password of multiple roles (if applicable)
- Provide clear screen shots of each page, explain the functions of each button

As a rough guide, the user guide should include the following sections:

System Document

In this section, students should provide the following pieces of information:

- **System (hardware and software requirements)**. Students should describe the minimum hardware and software requirements to install the software application which has been developed by the students, for example, DBMS, OS, program development tools etc.
- **Installation**. Under this section, students should create an 'installation' CD and provide a brief step-by-step guide on how a new user can install the software on a computer system. Students should indicate any special setup information, such as the specific location of placing the database files, the SQL statements to add tables, etc. Software source code should also be included in this CD. Students are not required to print out the software code.

Operation Document

Under this section, students are required to provide a brief step-by-step guide on how to use the installed software. The guides should teach the user how to run the software and use its major functions and features. For example, steps show guide the users on how to run the system, e.g. to use an executable file or to use the IDE. The login information such as username and password for each of the different users (or roles) must be provided. Some screen interfaces would be useful.

APPENDIX *n+1* Developer Guide*

*To be included for ***Smart Campus Projects*** and ***Real-Life Projects***. This section should include the following:

- List the necessary software, installer, API, library that must be installed
- List the authentication details, such as username and password for all security

Other Appendices:

- Supporting documents
- Non-disclosure agreement (if applicable)
- Other detailed documentation, such as important references, interview results, survey results, etc.